

University estates and asset-management teams

Stakeholder question catalogue

80 practical questions for traceable asset information, technically disciplined evidence, bounded analysis, assurance-aware decisions and long-term portfolio stewardship.

80

QUESTIONS

8

DECISION CATEGORIES

6

SUBSTANTIVE FIELDS PER RECORD

Purpose and intended use

This catalogue supports questions; it does not make decisions for accountable owners. It helps estates and asset-management teams locate evidence, understand uncertainty and identify the next accountable action across asset information, operations, projects, assurance, lifecycle planning and portfolio strategy.

An answer may be verified, conditional, conflicted, stale, unavailable or not authorised. Unknown is a valid and often safer outcome.

Verification gate for proposed evidence

- 1 Deployment**
The sensor, integration or analytical component exists in the intended scope.
- 2 Mapping**
Asset, point, meter, room and system identities are traceable, not merely probable.
- 3 Commissioning**
Units, intervals, calibration, expected ranges and failure modes are tested.
- 4 Permission**
Purpose, access, retention, privacy, cybersecurity and disclosure are authorised.
- 5 Current health**
Coverage, clock, communication, drift and data-quality status are current.

May 2021 floor plan: historical spatial reference only

The May 2021 floor plan may support historical room geometry or former labels. It cannot establish current room use, access, services, hazards, closures, tenure, capacity or maintenance routes. Every current spatial claim requires a controlled current source or competent field confirmation.

The service must not

Approve investment, procurement, disposal or maintenance; certify condition, compliance, completion, capacity or safety; create valuations; issue work, access, switching or isolation authority; control plant; infer people, productivity, legal duty or research activity; disclose restricted asset information; or replace surveys, finance records, competent professionals and accountable asset owners.

Evidence precedence

Current authoritative owners and competent professionals always take precedence. Source precedence is set for the decision context; it is not inferred from file age or system prestige. Every answer preserves effective/observed and retrieval times, provenance, permission, coverage, staleness and conflicts. When evidence cannot support the claim, the answer stops.

Field guide and analytical operation types

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Every record contains six substantive fields and an explicit safe-failure rule

1

Stakeholder question

A natural decision question a university estates or asset-management team would genuinely ask.

2

Sensors and telemetry

Proposed asset-level or aggregate evidence, with modality, location, coverage, timing, quality and deployment conditions.

3

Authoritative supporting data

Named current owners, precedence, effective/observed and retrieval times, permissions, provenance, staleness and conflicts.

4

Analysis required

Operation, decision cadence, minimum evidence, method, uncertainty and the result when evidence is insufficient.

5

Answer boundary

What the service must not infer, disclose, promise, certify, authorise or control.

6

Why this is useful

The concrete decision, verification step, work queue or accountable route enabled by the answer.

Operation labels used in the catalogue

Lookup

Retrieve a controlled fact without changing it.

Observation

Report measured or directly recorded evidence within its verified scope.

Calculation

Apply a stated arithmetic or accounting method to eligible inputs.

Estimate

Quantify a present unknown as a range with assumptions and uncertainty.

Forecast

Project a future range under a declared scenario; it is not a promise.

Comparison

Test differences, consistency or thresholds without silently reconciling conflicts.

Diagnosis

Identify plausible patterns or candidates; competent investigation confirms cause.

Recommendation

Rank or route options; accountable owners retain decision and authorisation.

Catalogue map and common evidence rules

Eight categories, ten questions in each

Question categories

01 asset registers, hierarchy and data quality

Questions 1-10 | Trustworthy identity, hierarchy, location, attributes and correction priorities.

02 condition, criticality, risk and backlog

Questions 11-20 | Condition evidence, consequence-led priorities and transparent backlog decisions.

03 maintenance strategy, work history and reliability

Questions 21-30 | Planned and reactive maintenance evidence, work quality, failure patterns and reliability.

04 space, tenure, ownership and operational use

Questions 31-40 | Current space facts, tenure and ownership authority, operational use and access boundaries.

05 projects, changes, handover and information management

Questions 41-50 | Coordinated change, controlled handover, information completeness and accountable acceptance.

06 warranties, statutory records and assurance boundaries

Questions 51-60 | Warranty and statutory evidence, due actions, assurance status and authority boundaries.

07 cost, value, lifecycle and investment evidence

Questions 61-70 | Cost evidence, lifecycle ranges, option comparison and accountable investment decisions.

08 portfolio resilience, decarbonisation and scenario planning

Questions 71-80 | Resilience, decarbonisation pathways, dependencies, scenarios and evidence gaps.

Common evidence rules

1. Scope before signal

Define asset class, building, system, served area, decision date and permitted detail before retrieval.

2. Current authority

Use the named owner and controlled source for the decision. Corroboration may raise an exception but cannot silently replace authority.

3. Time is part of the evidence

Carry effective or observed time, retrieval time, interval and clock-alignment quality into the answer.

4. Quality is decision-specific

Check mapping, calibration or functional verification, uptime, gaps, drift, coverage and provenance against the intended decision.

5. Conflict is an answer state

Show material conflicts and their owners. Do not average, overwrite or choose a convenient value.

6. Minimum evidence or stop

If minimum evidence is absent, return unavailable, conflicted, stale or verification required rather than an unsupported claim.

7. Privacy and security by minimisation

Use the least detail needed; suppress low-count or restricted location outputs; do not identify or track people.

8. Accountable action remains human

The service can assemble, compare, calculate, estimate, forecast, diagnose candidates or recommend routes. It does not authorise outcomes.

EA-001	STAKEHOLDER QUESTION		
	For the agreed maintainable scope, which installed assets are missing from the live register, and which registered assets cannot be verified on site?		
	SENSORS AND TELEMETRY	AUTHORITATIVE SUPPORTING DATA	ANALYSIS REQUIRED
	Proposed evidence: commissioned QR or barcode verification, controller inventories, gateway device lists and non-identifying equipment-state heartbeats. Coverage: accessible plant, distribution assets and maintainable room equipment included in the agreed survey scope; no inference behind closed fabric. Timing: at mobilisation, after a project or move, and then by risk-based sample each quarter; aggregate by building, system and asset class. Quality limits: untagged items, inaccessible rooms, duplicate network addresses and offline gateways create false absences. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.	Primary authority: the current CAFM or EAM asset register and approved asset scope rules, maintained by the Estates Asset Information owner. Corroboration: approved as-built schedules, O&M manuals, BIM or CDE objects, project handover lists and current maintainer inventories. Precedence: the current approved register governs identity and status; observed equipment may raise an exception but cannot create a record automatically. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag records beyond the owner-defined refresh period, unmatched handover rows and source scope differences. Preserve unresolved conflicts for the accountable owner; never select a convenient value.	Operation: comparison and lookup. Decision cadence: expected-versus-observed completeness checking, monthly and after material change. Minimum evidence: a declared asset-class scope, at least one stable identifier or serial attribute, and location or system evidence. Method: rule-based coverage checks followed by probabilistic entity matching for near matches. Uncertainty: coverage percentage, unmatched counts and confidence bands by class. Safe failure: return a verification list with reasons and locations; do not claim completeness where access or class scope is unknown.
EA-002	STAKEHOLDER QUESTION		
	Which records are credible candidates for the same physical asset, and what evidence separates a duplicate from a replacement or a different unit?		
	SENSORS AND TELEMETRY	AUTHORITATIVE SUPPORTING DATA	ANALYSIS REQUIRED
	Proposed evidence: tag scans, serial-number capture, controller object identifiers and short, permissioned equipment observations such as make, model and rated duty. Coverage: candidate duplicates within the same building or connected system, excluding personal devices and research equipment unless explicitly in scope. Timing: on data load, at handover and when a work order reveals a mismatch; aggregate as candidate pairs, never as a silently merged record. Quality limits: replacement plates, reused tags, transcription errors and identical packaged units can make separate assets look alike. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.	Primary authority: current asset master records, maintained by the Estates Asset Information owner. Corroboration: purchase and disposal records, project schedules, warranty registrations, O&M files and maintainer notes. Precedence: only the accountable asset owner may approve a merge or supersession; signed disposal or replacement evidence overrides similarity. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag missing serials, reused local names, stale locations and conflicting installation dates. Preserve unresolved conflicts for the accountable owner; never select a convenient value.	Operation: comparison and entity resolution. Decision cadence: duplicate detection and entity resolution, on each import and weekly for new or changed records. Minimum evidence: two independent matching attributes, or one unique serial plus credible location and status evidence. Method: deterministic rules for exact keys, then weighted similarity across identifiers, dates, models, parents and locations. Uncertainty: pairwise confidence and the attributes driving the match. Safe failure: present possible pairs for human decision; keep records separate when evidence is ambiguous.

EA-003

STAKEHOLDER QUESTION

Is each asset linked to the correct parent, engineering system, served area and service dependency?

SENSORS AND TELEMETRY

Proposed evidence: commissioned BMS point-to-equipment mappings, electrical circuit monitoring, valve or damper position where already instrumented, and tagged walk-down confirmation. Coverage: assets whose parent, served area or system relationship is in the approved information scope. Timing: after commissioning, configuration change or repeated relationship exception; otherwise quarterly; aggregate to system topology and room or zone level, not individual occupant level. Quality limits: shared services, temporary supplies, undocumented cross-connections and point-name drift can misstate relationships. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: approved asset hierarchy and system schematics, maintained by the Estates Asset Information owner with the relevant engineering system owner. Corroboration: single-line diagrams, valve schedules, panel schedules, BIM relationships, BMS cause-and-effect descriptions and work history. Precedence: current approved engineering schematics and accountable owner confirmation take precedence over inferred telemetry relationships. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag superseded drawings, unapproved mark-ups, incomplete served-area lists. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: comparison and bounded diagnosis. Decision cadence: hierarchy and topology validation, quarterly and whenever a project changes a service path. Minimum evidence: a child identifier, proposed parent or system, location, and at least one independent connectivity or document link. Method: graph consistency rules plus correlation checks on state changes only where controlled tests or natural events provide evidence. Uncertainty: relationship confidence and alternative parent candidates. Safe failure: mark the relationship unverified and request an engineering walk-down when connectivity cannot be distinguished safely.

ANSWER BOUNDARY

Do not command equipment to create test events, infer hidden pipe or cable routes, or overwrite approved engineering relationships. The accountable asset-information owner retains authority over record creation, change and retirement.

WHY THIS IS USEFUL

The relevant system owner can target topology validation before isolation, impact or cost roll-up depends on the relationship.

EA-004

STAKEHOLDER QUESTION

Which maintainable assets lack a stable identifier, readable label or scannable tag at the point of work?

SENSORS AND TELEMETRY

Proposed evidence: permissioned tag scans, handheld barcode readability checks and close-range label-quality observations recorded without people in frame. Coverage: maintainable assets in accessible areas and agreed plant spaces. Timing: during planned inspections, handover and first attendance on a work order; aggregate counts and exception lists by asset class and location. Quality limits: damaged labels, unsafe access, condensation, glare and duplicate labels can make scans unreliable. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: asset naming and tagging standard plus current register identifiers, maintained by the Estates Asset Information owner. Corroboration: project tagging schedules, commissioning sheets, manufacturer plates and maintainer exceptions. Precedence: the owner-approved identifier remains authoritative; physical labels are evidence of implementation, not authority to rename. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag temporary labels, legacy naming schemes, missing installation dates and duplicate tag assignments. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: observation and comparison. Decision cadence: identifier coverage and uniqueness checking, monthly until stable, then quarterly. Minimum evidence: an in-scope asset list, expected tag format and one on-site observation per exception. Method: format validation, duplicate-key checks and readability scoring. Uncertainty: verified, unreadable, absent, duplicate or access-not-obtained states. Safe failure: return an exception without suggesting a new identifier where naming authority or asset identity is unclear.

ANSWER BOUNDARY

Do not assign identifiers, print tags, expose security-sensitive labels or require unsafe access to read a plate. The accountable asset-information owner retains authority over record creation, change and retirement.

WHY THIS IS USEFUL

The team can plan a safe tagging campaign and prevent work history being attached to the wrong item.

EA-005

STAKEHOLDER QUESTION

Which decision-critical asset attributes are missing, stale or contradictory, including make, model, duty, capacity, installation date and service owner?

SENSORS AND TELEMETRY

Proposed evidence: runtime counters, controller firmware and configuration reads, metered capacity observations and tagged field verification where technically and contractually permitted. Coverage: attributes needed for maintenance, risk, energy or investment decisions; exclude unrelated device content. Timing: on retrieval for a decision, after change and by owner-defined refresh cycle; aggregate by attribute, source and decision consequence. Quality limits: sensor calibration, controller replacements, estimated values and inaccessible plates limit confidence. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the current asset master and attribute data dictionary, maintained by the Estates Asset Information owner. Corroboration: approved O&M data, commissioning records, manufacturer documentation, work orders, finance records and engineering surveys. Precedence: the named authoritative source for each attribute wins unless the accountable owner records an exception; observed data may challenge but not replace it. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag values older than the attribute-specific tolerance, contradictory units, missing provenance and estimates presented as measured. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: lookup and comparison. Decision cadence: attribute quality scoring and decision-impact assessment, at each material decision and monthly for high-impact fields. Minimum evidence: field definition, owner, accepted units, timestamp, provenance and at least one comparison source. Method: validity, completeness, consistency and timeliness rules weighted by the decisions that use the field. Uncertainty: a per-field confidence grade and stated consequence of uncertainty. Safe failure: withhold the affected calculation and state which attribute needs confirmation when minimum quality is not met.

ANSWER BOUNDARY

Do not invent missing values, convert estimates into facts, disclose restricted technical attributes or certify that a record is correct. The accountable asset-information owner retains authority over record creation, change and retirement.

WHY THIS IS USEFUL

The data owner can prioritise only the attributes capable of changing the pending maintenance, risk or lifecycle decision.

EA-006

STAKEHOLDER QUESTION

Does the recorded location match where each asset can currently be found at the precision needed for maintenance and risk decisions?

SENSORS AND TELEMETRY

Proposed evidence: scannable tag observations, permissioned indoor survey coordinates, plant-room gateway proximity and controller-to-zone mappings. Coverage: accessible assets with a location precision appropriate to safe finding, such as building, floor, room or plant zone. Timing: at handover, after moves and when first attendance fails to find an asset; aggregate to the least-sensitive useful location level. Quality limits: radio proximity is coarse, mobile surveys drift indoors, and a historical floor plan cannot prove current access or use. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the current space hierarchy and asset location fields, maintained by the Estates Space Information and Asset Information owners. Corroboration: approved current plans, project records, room signage records, BMS graphics and maintainer observations. Precedence: current approved space identifiers and accountable owner confirmation take precedence; the May 2021 floor plan is historical reference only. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag renumbered rooms, subdivided spaces, moved assets, restricted areas and conflicting plan versions. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: observation and comparison. Decision cadence: location reconciliation, on exception and quarterly for critical assets. Minimum evidence: asset identifier, current space identifier, observation time and an independent location source. Method: exact hierarchy checks plus bounded proximity matching where survey accuracy is known. Uncertainty: location precision, observation age and match confidence. Safe failure: report the last verified location and request a safe site check when access, mapping or precision is insufficient.

ANSWER BOUNDARY

Do not expose sensitive plant locations to unauthorised users, infer access rights, track people or direct a user into a closed or hazardous area. The accountable asset-information owner retains authority over record creation, change and retirement.

WHY THIS IS USEFUL

Planners can request location verification before attendance and reduce failed visits or unsafe searching.

EA-007

STAKEHOLDER QUESTION

Which building-services sensors, controllers and gateways should be managed as estate assets but have no agreed place in the asset hierarchy?

SENSORS AND TELEMETRY

Proposed evidence: network-management inventory limited to approved building-control segments, gateway heartbeats, controller point lists and physical tag verification. Coverage: building-service devices under Estates responsibility; explicitly exclude personal, teaching, research and general IT endpoints unless governance assigns them. Timing: weekly discovery comparison and at commissioning or decommissioning; aggregate by control system, plant area and device class. Quality limits: MAC changes, virtual points, dormant devices, vendor gateways and inaccessible panels can create false positives. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the approved control-system asset scope and current asset register, maintained by the Estates controls owner and Asset Information owner. Corroboration: BMS architecture, cyber asset inventory, vendor support records, project handover and maintenance contracts. Precedence: governed scope and accountable ownership determine whether a device is an estate asset; discovery alone has no precedence. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag unowned devices, obsolete point lists, duplicate virtual instances and contract boundaries. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: lookup and classification. Decision cadence: scope classification and discovered-versus-recorded reconciliation, weekly for connected controls and after system change. Minimum evidence: device identity, segment or panel context, declared owner, function and one physical or handover corroborator. Method: allow-list classification, exact and fuzzy matching, then exception clustering by system. Uncertainty: in-scope probability and reasons for exclusion or ambiguity. Safe failure: quarantine the finding as an information exception and escalate ownership; do not enrol it automatically.

ANSWER BOUNDARY

Do not scan unapproved networks, inspect payload content, expose cyber-sensitive topology or assert that every reachable device is safe, supported or owned by Estates. The accountable asset-information owner retains authority over record creation, change and retirement.

WHY THIS IS USEFUL

The service owner can identify unmanaged controls that need ownership, support and lifecycle treatment.

EA-008

STAKEHOLDER QUESTION

Which assets appear active in one system but isolated, replaced, removed or out of service in another?

SENSORS AND TELEMETRY

Proposed evidence: long-term absence of commissioned heartbeats, zero runtime with corroborating isolation state, tag verification and replacement-device commissioning evidence. Coverage: assets with expected telemetry or accessible physical confirmation; never infer disposal from silence alone. Timing: weekly exception detection with confirmation after project or maintenance activity; aggregate by asset status, system and responsible owner. Quality limits: communications outages, seasonal shutdown, standby duty and telemetry retirement can mimic decommissioning. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: current asset status and change-control records, maintained by the Estates Asset Information owner. Corroboration: project completion records, disposal notes, work orders, finance asset records, BMS configuration history and warranty registrations. Precedence: signed change, disposal or replacement records and accountable owner confirmation take precedence over inactivity. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag open project records, missing disposal evidence, unclosed work orders and conflicting finance or operational status. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: comparison. Decision cadence: status anomaly detection, weekly with owner confirmation before any action. Minimum evidence: previous active state, evidence of a successor or approved change, and at least one physical, project or finance corroborator. Method: state-transition rules and temporal anomaly checks across work, telemetry and records. Uncertainty: possible moved, replaced, dormant, telemetry-failed or decommissioned states with confidence. Safe failure: retain the current status and create a verification task when evidence cannot distinguish these states.

ANSWER BOUNDARY

Do not retire assets, stop maintenance, write off value or infer disposal solely from missing telemetry. The accountable asset-information owner retains authority over record creation, change and retirement.

WHY THIS IS USEFUL

The accountable owner can verify status before stopping work, retiring records or changing financial treatment.

EA-009

STAKEHOLDER QUESTION

Can every asset fact used in this answer be traced to a current source owner, evidence item and time?

SENSORS AND TELEMETRY

Proposed evidence: signed survey observations, tag scans and telemetry readings carrying device identity, calibration or health state, location scope and timestamp. Coverage: decision-relevant attributes for critical or high-value assets. Timing: whenever a value is created or used, with daily provenance integrity checks; aggregate as an attribute-level evidence chain rather than a single record score. Quality limits: manual transcription, clock drift, data transformations and vendor exports may break lineage. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the asset data dictionary, source register and ownership matrix, maintained by the Estates information governance owner. Corroboration: CAFM or EAM history, CDE document metadata, work orders, finance extracts, sensor metadata and access logs. Precedence: the named authoritative owner and approved source mapping govern each field; copied values never outrank their origin. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag orphaned values, missing effective dates, transformations without method and conflicting owner claims. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: lookup and provenance validation. Decision cadence: provenance-chain validation, on data publication and before high-consequence use. Minimum evidence: source identifier, owner role, effective or observation time, retrieval time, transformation history and permission label. Method: lineage graph traversal with required-field and broken-link checks. Uncertainty: complete, partial or broken lineage plus the exact missing links. Safe failure: withhold the unsupported fact from decision calculations and show the last traceable source.

ANSWER BOUNDARY

Do not reveal restricted sources, imply that provenance proves correctness, or nominate a responsible owner without an approved governance record. The accountable asset-information owner retains authority over record creation, change and retirement.

WHY THIS IS USEFUL

Decision-makers can see which facts are traceable, which are disputed and where owner confirmation is required.

EA-010

STAKEHOLDER QUESTION

Which asset-data defects should be corrected first because they could affect an imminent operational, financial or project decision?

SENSORS AND TELEMETRY

Proposed evidence: data-quality exceptions from commissioned asset observations, control-system mappings and tagged verification. Coverage: all in-scope asset classes, summarised without exposing restricted locations or security-sensitive details. Timing: monthly and before major planning cycles; aggregate by issue type, asset criticality, decision process and accountable data owner. Quality limits: instrumented assets are over-represented, so lack of telemetry must not be mistaken for low consequence. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the current asset-data issue register and decision dependency map, maintained by the Estates information governance owner. Corroboration: criticality records, maintenance plans, work failures, project pipelines, finance models and assurance schedules. Precedence: current accountable owners determine consequence and remediation priority; automated scores are advisory. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag old issue severity, duplicate tickets, missing impact owners and conflicts between operational and finance priorities. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: recommendation from comparison. Decision cadence: risk-based data-remediation prioritisation, monthly with event-triggered reprioritisation. Minimum evidence: a defined issue, affected decisions, asset or system scope, consequence owner and evidence of recurrence or exposure. Method: multi-criteria ranking using consequence, likelihood, reach, urgency, effort and dependency, with sensitivity testing. Uncertainty: score ranges and rank stability under different weights. Safe failure: present an unranked escalation set when owners or consequence assumptions are missing.

ANSWER BOUNDARY

Do not authorise remediation spend, assign blame, alter ownership or hide low-confidence issues behind a composite score. The accountable asset-information owner retains authority over record creation, change and retirement.

WHY THIS IS USEFUL

The information team can direct limited cleansing capacity to defects with the highest near-term consequence.

EA-011

STAKEHOLDER QUESTION

Which assets show credible evidence of deteriorating condition, and which apparent trends may instead reflect duty, season or sensor behaviour?

SENSORS AND TELEMETRY

Proposed evidence: trendable vibration, temperature, differential pressure, electrical current, leak detection, run-hour and alarm-state data selected for the specific asset class. Coverage: critical or failure-prone assets with an approved sensing plan; not every condition mode is remotely observable. Timing: continuous or periodic sampling at engineering-appropriate rates, summarised daily and compared monthly; aggregate per asset and comparable duty state. Quality limits: load changes, seasonal operation, sensor drift and maintenance resets can look like deterioration. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: current condition records and defect history, maintained by the Estates condition and maintenance owners. Corroboration: commissioning baselines, work orders, operator inspections, specialist surveys, manufacturer limits and BMS trend logs. Precedence: competent condition assessments and approved engineering limits take precedence over anomaly scores. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag expired surveys, changed duty, missing calibration, inconsistent condition scales. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: observation and bounded diagnosis. Decision cadence: condition trend detection, daily screening with monthly engineering interpretation. Minimum evidence: a stable asset identity, known duty state, healthy sensors, a baseline and repeated observations across comparable operation. Method: change-point and trend analysis normalised for load and season, checked against known interventions. Uncertainty: trend magnitude, confidence interval and alternative explanations. Safe failure: state that deterioration is unconfirmed and request targeted inspection when evidence is sparse, non-comparable or sensor health is poor.

ANSWER BOUNDARY

Do not certify condition, predict an exact failure date, command plant, or replace a competent inspection. The answer supports triage only; competent professionals and accountable owners retain condition, risk and backlog decisions.

WHY THIS IS USEFUL

Condition teams can target inspection at assets with credible change while separating seasonal or duty effects.

EA-012

STAKEHOLDER QUESTION

Are current criticality classifications still aligned with the services, academic calendar and recovery needs each asset supports?

SENSORS AND TELEMETRY

Proposed evidence: service-state, availability, dependency and aggregate demand telemetry from commissioned building systems. Coverage: systems supporting declared critical activities; occupancy or demand data must be non-identifying and aggregated. Timing: near real time for status, with termly or annual criticality reassessment; aggregate by service, building and time window. Quality limits: telemetry shows use and dependency, not institutional priority or consequence by itself. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the approved asset criticality framework and business-service register, maintained by the accountable Estates asset owners with service owners and institutional risk owners. Corroboration: business continuity plans, teaching timetables, research support requirements, hazard records, service-level commitments and dependency diagrams. Precedence: approved service-owner consequence statements and institutional risk governance take precedence over utilisation metrics. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag unconfirmed service dependencies, changed teaching or research needs, missing recovery objectives and conflicting stakeholder priorities. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: comparison and recommendation. Decision cadence: multi-domain criticality assessment, annually and after material service or dependency change. Minimum evidence: asset-to-service mapping, consequence categories, recovery requirement, redundancy evidence and accountable owner confirmation. Method: scored consequence and redundancy assessment with scenario walk-through and sensitivity to disputed weights. Uncertainty: criticality band, confidence and domains driving the result. Safe failure: leave criticality unassigned or provisional where consequence ownership or dependencies are not confirmed.

ANSWER BOUNDARY

Do not infer research importance, rank people or departments, disclose sensitive research dependencies, or set institutional risk appetite. The answer supports triage only; competent professionals and accountable owners retain condition, risk and backlog decisions.

WHY THIS IS USEFUL

Owners can recheck contingency, maintenance and spares priorities against current university service needs.

EA-013

STAKEHOLDER QUESTION

Where do condition grades, criticality classes, risk scores and backlog priorities contradict one another?

SENSORS AND TELEMETRY

Proposed evidence: asset-condition indicators and service availability signals already approved for the relevant equipment. Coverage: assets carrying all three governance fields, with gaps shown rather than imputed. Timing: monthly and before backlog or capital prioritisation; aggregate by asset, system and contradiction type. Quality limits: different assessment dates and definitions can create apparent contradictions. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: approved condition, criticality and risk records, maintained by the respective Estates condition, asset and risk owners. Corroboration: survey reports, business continuity evidence, defect records, mitigations and work history. Precedence: each owner retains authority for its field; the service must compare definitions and dates rather than choose a winner. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag scores based on superseded evidence, changed scales, missing mitigations and conflicting effective dates. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: comparison. Decision cadence: cross-register consistency analysis, monthly and at decision gates. Minimum evidence: definitions, assessment dates, source evidence, current mitigations and the accountable owner for each score. Method: rule-based contradiction patterns such as poor condition with low risk, or high criticality with no contingency, followed by owner validation. Uncertainty: type and severity of contradiction plus freshness-adjusted confidence. Safe failure: show the conflict and exclude it from automated ranking until owners reconcile it.

ANSWER BOUNDARY

Do not overwrite scores, decide risk tolerance, certify condition or present a single blended number as authoritative. The answer supports triage only; competent professionals and accountable owners retain condition, risk and backlog decisions.

WHY THIS IS USEFUL

The accountable owners can reconcile incompatible ratings before they drive backlog or capital prioritisation.

EA-014

STAKEHOLDER QUESTION

Which new observations justify a defect candidate for the backlog, and what evidence is still needed before priority can be set?

SENSORS AND TELEMETRY

Proposed evidence: leak, temperature, vibration, electrical protection, alarm and manual mobile-inspection observations appropriate to the asset. Coverage: agreed assets and spaces with privacy-preserving observations; no general surveillance. Timing: event-driven, with persistence checks over minutes, hours or operating cycles as appropriate; aggregate as defect candidates linked to asset and location. Quality limits: single alarms, sensor faults, transient operating states and duplicate reports can create false defects. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the defect and backlog entry criteria, maintained by the Estates maintenance and condition owners. Corroboration: active work orders, asset criticality, known hazards, operator reports, manufacturer limits and recent condition evidence. Precedence: authorised defect triage and competent safety assessment take precedence over an automated candidate. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag duplicate tickets, unclosed related work, missing asset identity and stale criticality. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: observation and recommendation. Decision cadence: defect candidate detection and triage support, at event time with human triage in the required operational window. Minimum evidence: persistent or corroborated evidence, asset and location, current operating context and a check for existing work. Method: thresholds, state rules and duplicate detection, with consequence prompts based on approved criticality. Uncertainty: candidate confidence and evidence completeness, not a final priority. Safe failure: raise an information or safety escalation and avoid backlog classification when asset, consequence or sensor health is uncertain.

ANSWER BOUNDARY

Do not create work orders, set final priority, declare an area safe, diagnose a root cause or authorise isolation. The answer supports triage only; competent professionals and accountable owners retain condition, risk and backlog decisions.

WHY THIS IS USEFUL

Triage staff can route a well-evidenced defect candidate and identify what is missing before priority is assigned.

EA-015

STAKEHOLDER QUESTION

Which backlog items have exceeded their target, deferral condition or current risk tolerance, and what has changed since the last authorised decision?

SENSORS AND TELEMETRY

Proposed evidence: current service-state and temporary-mitigation status from commissioned systems, plus tagged verification where a backlog item depends on physical state. Coverage: open items with an asset or space link; aggregate sensitive locations for wider reporting. Timing: daily ageing calculation and event updates when condition or mitigation changes; aggregate by priority, owner, system and reason for delay. Quality limits: open telemetry cannot prove that a defect remains, and silent telemetry cannot prove resolution. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the authoritative backlog and target-date records, maintained by the Estates maintenance planning owner. Corroboration: risk acceptances, temporary controls, access constraints, project dependencies, purchase orders and work evidence. Precedence: approved priority, target and risk-acceptance records govern ageing; current competent safety decisions override schedule logic. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag paused items without reasons, changed targets, duplicate items, expired risk acceptances and missing closure evidence. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: lookup and comparison. Decision cadence: backlog ageing and tolerance exception analysis, daily with weekly management aggregation. Minimum evidence: open status, approved priority, target date, current mitigation, owner and latest evidence. Method: calendar and working-day ageing, pause-reason validation, and re-exposure checks when mitigations expire. Uncertainty: days beyond target, confidence in open state and mitigation expiry risk. Safe failure: show an exception without escalating priority automatically when status or approved tolerance is unclear.

ANSWER BOUNDARY

Do not change dates, priorities or risk acceptances, infer negligence, or certify that a mitigation remains effective. The answer supports triage only; competent professionals and accountable owners retain condition, risk and backlog decisions.

WHY THIS IS USEFUL

Owners can escalate overdue exposure, retest deferral assumptions and confirm whether temporary controls remain valid.

EA-016

STAKEHOLDER QUESTION

Which assets have repeated temporary interventions, bypasses or symptom recurrence without evidence of a durable resolution?

SENSORS AND TELEMETRY

Proposed evidence: repeated reset, override, bypass, alarm-clear and short-lived performance-recovery events from approved control systems. Coverage: assets where these states are safe and meaningful to monitor; exclude user actions unrelated to plant management. Timing: event-level capture with weekly pattern summaries; aggregate per asset and failure mode. Quality limits: planned testing, commissioning and seasonal changes can resemble temporary intervention. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: work-order actions and defect history, maintained by the Estates maintenance owner. Corroboration: BMS change logs, permits, approved temporary measures, contractor reports, spares constraints and project plans. Precedence: authorised work records and competent engineering interpretation take precedence over event patterns. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag generic closure notes, missing cause codes, undocumented overrides and interventions recorded against parent assets. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: bounded diagnosis. Decision cadence: recurrence and temporary-fix pattern analysis, weekly and before maintenance planning. Minimum evidence: at least two comparable events or interventions, linked asset identity, timestamps and evidence of subsequent symptom return. Method: sequence mining and recurrence windows, separating planned tests and approved temporary controls. Uncertainty: recurrence count, duration of relief and classification confidence. Safe failure: label the pattern for investigation rather than declaring the intervention ineffective when work evidence is incomplete.

ANSWER BOUNDARY

Do not diagnose root cause, remove overrides, authorise permanent work or criticise individual contractors. The answer supports triage only; competent professionals and accountable owners retain condition, risk and backlog decisions.

WHY THIS IS USEFUL

Maintenance leads can commission root-cause investigation where repeated effort has not restored stable service.

EA-017

STAKEHOLDER QUESTION

Which critical services depend on a single asset, path or control point without a verified and operable contingency?

SENSORS AND TELEMETRY

Proposed evidence: service availability, switch position, pump or fan duty, networked control state and periodic failover-test evidence where approved. Coverage: declared critical services and their supporting plant; no active test is initiated by the service. Timing: near real time for status and after each approved failover test or topology change; aggregate as dependency paths and service-level redundancy state. Quality limits: standby presence does not prove capacity, fuel, control logic or successful changeover. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: approved system topology and business continuity arrangements, maintained by the accountable engineering and business continuity owners. Corroboration: single-line diagrams, cause-and-effect records, test reports, generator or UPS records, fuel plans, maintenance status and critical-service mappings. Precedence: current approved topology and competent test evidence take precedence; telemetry only reflects observed state. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag untested standby assets, stale capacity assumptions, temporary isolations and undocumented cross-connections. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: comparison and estimate. Decision cadence: single-point-of-failure and contingency verification analysis, monthly and after any topology or critical-service change. Minimum evidence: complete dependency path, asset availability, capacity requirement, contingency asset or route and recent test evidence. Method: graph cut-set analysis plus capacity and test-validity checks under defined scenarios. Uncertainty: verified, partial or unverified resilience with scenario-specific confidence. Safe failure: return an unresolved dependency and request engineering validation where the topology or contingency capacity is incomplete.

ANSWER BOUNDARY

Do not initiate failover, promise continuity, disclose sensitive resilience details broadly or certify emergency arrangements. The answer supports triage only; competent professionals and accountable owners retain condition, risk and backlog decisions.

WHY THIS IS USEFUL

Resilience owners can target contingency verification, spares or redundancy work at genuine single points of dependency.

EA-018

STAKEHOLDER QUESTION

Which condition grades or survey findings are too old, narrow or weakly evidenced for the decisions now being made?

SENSORS AND TELEMETRY

Proposed evidence: current operating indicators and tagged observations relevant to the last assessed condition mode. Coverage: assets carrying a condition grade, prioritised by criticality and time since assessment. Timing: monthly screening and before each investment or disposal decision; aggregate by grade age, evidence type and asset class. Quality limits: stable telemetry cannot rule out hidden fabric, corrosion, wear or statutory defects. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the condition assessment record and assessment standard, maintained by the Estates condition owner. Corroboration: surveyor reports, photographs with controlled access, inspection scope, work completed since assessment and manufacturer information. Precedence: the latest competent assessment within its stated scope governs; sensor trends may trigger reassessment but cannot extend validity. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag grades beyond class-specific validity, missing scope, assessor or date, post-assessment changes and conflicts with recent defects. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: lookup and comparison. Decision cadence: condition-evidence validity checking, monthly and at decision gate. Minimum evidence: grade, assessment date, method, scope, assessor competence evidence, intervening changes and current criticality. Method: age thresholds by asset class, scope completeness tests and event-based invalidation rules. Uncertainty: usable, provisional or unsuitable status with reasons. Safe failure: withhold condition-based calculations and recommend a competent assessment when validity cannot be established.

ANSWER BOUNDARY

Do not renew a grade, certify condition, infer hidden condition from absence of alarms or publish restricted survey images. The answer supports triage only; competent professionals and accountable owners retain condition, risk and backlog decisions.

WHY THIS IS USEFUL

Survey budgets can be directed to evidence that is too weak or old for the current decision.

EA-019

STAKEHOLDER QUESTION

What is the plausible service consequence if this asset fails during the next defined teaching, research or campus operating window?

SENSORS AND TELEMETRY

Proposed evidence: current load, run state, aggregate demand, standby availability and environmental or process response signals. Coverage: the asset, served zones and declared service window; use aggregate, non-identifying demand only. Timing: on request using current status and representative historical windows; aggregate by service consequence and affected time interval. Quality limits: future demand, human responses, research schedules and cascading failures may be uncertain. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: asset-to-service dependencies and approved operating schedules, maintained by the accountable asset and service owners. Corroboration: teaching timetables, event bookings, critical research requirements, continuity plans, redundancy tests, access constraints and current work. Precedence: authorised service schedules and owner consequence statements take precedence over historical occupancy patterns. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag late schedule changes, unverified dependencies, expired contingency tests and restricted service information. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: estimate and forecast. Decision cadence: bounded failure-impact scenario, for the next shift, day, week or defined operational window. Minimum evidence: confirmed dependency, current asset state, expected service demand, redundancy status and consequence owner. Method: deterministic dependency tracing with scenario ranges for demand and restoration time. Uncertainty: best-supported consequence range and named unknowns. Safe failure: state that impact is indeterminate and escalate to service and engineering owners if dependencies or contingency are not verified.

ANSWER BOUNDARY

Do not predict exact people affected, promise restoration, declare continued operation safe, authorise shutdown or disclose sensitive research activity. The answer supports triage only; competent professionals and accountable owners retain condition, risk and backlog decisions.

WHY THIS IS USEFUL

Planners can select a safer intervention window and prepare proportionate service-continuity arrangements.

EA-020

STAKEHOLDER QUESTION

Which assets or areas need a competent on-site survey because available condition, risk or dependency evidence is insufficient?

SENSORS AND TELEMETRY

Proposed evidence: sensor-health flags, contradictory condition indicators, unresolved tag or location exceptions and inaccessible-asset status. Coverage: assets and areas within the decision scope, with restricted locations visible only to authorised roles. Timing: weekly exception aggregation and before high-consequence decisions; aggregate by survey need, discipline and urgency evidence. Quality limits: lack of telemetry is common and does not by itself indicate poor condition. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: current surveys, defects, risk records and access constraints, maintained by the Estates condition owner and relevant competent professional. Corroboration: asset register, work history, project changes, assurance records, user reports and approved hazard information. Precedence: competent professionals decide survey scope and urgency; the service only assembles evidence gaps. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag expired surveys, changed use, unresolved conflicts, missing access information and decisions relying on unsupported assumptions. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: recommendation. Decision cadence: evidence sufficiency and survey-trigger assessment, weekly and at each decision gate. Minimum evidence: the intended decision, minimum evidence standard, latest competent assessment, current change history and known hazards. Method: gap analysis against decision-specific evidence requirements, weighted by consequence and irreducible uncertainty. Uncertainty: which evidence is missing and how strongly it affects the decision. Safe failure: recommend no remote conclusion and route to the appropriate competent discipline when the minimum is not met.

ANSWER BOUNDARY

Do not specify unsafe access, define professional scope beyond approved standards, certify urgency or replace a competent survey. The answer supports triage only; competent professionals and accountable owners retain condition, risk and backlog decisions.

WHY THIS IS USEFUL

The team can commission a competent, decision-specific survey instead of extrapolating from insufficient records.

EA-021

STAKEHOLDER QUESTION

Are planned maintenance tasks aligned with each asset type, duty, environment, criticality and governing obligation?

SENSORS AND TELEMETRY

Proposed evidence: run hours, starts, load bands, environmental exposure, alarm burden and standby duty from commissioned sources. Coverage: assets with an approved maintenance strategy and meaningful telemetry; no inference for uninstrumented failure modes. Timing: monthly utilisation summaries and before annual plan release; aggregate by asset class, duty profile and criticality band. Quality limits: changed duty, counter resets, missing seasonal operation and sensor drift can misclassify use. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the approved planned-maintenance library and asset maintenance plan, maintained by the Estates maintenance strategy owner. Corroboration: manufacturer instructions, warranty conditions, statutory schedules, risk assessments, work history and operating context. Precedence: statutory and warranty obligations plus competent engineering decisions take precedence over utilisation-based suggestions. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag generic task templates, wrong asset-class mappings, superseded instructions and unapproved interval changes. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: comparison. Decision cadence: maintenance-plan applicability checking, monthly and at annual planning. Minimum evidence: asset identity, task set, duty evidence, environment, criticality and mandatory obligations. Method: rule comparison to approved task standards, then bounded utilisation adjustment candidates. Uncertainty: compliance constraints, mismatch type and confidence. Safe failure: retain the approved task and flag for engineering decision when evidence or authority to vary it is absent.

ANSWER BOUNDARY

Do not change maintenance intervals, waive mandatory tasks, authorise work or present usage optimisation as compliance. The accountable maintenance owner retains authority over task, interval, closure and intervention decisions.

WHY THIS IS USEFUL

Maintenance owners can identify plans needing competent adjustment while protecting mandatory obligations.

EA-022

STAKEHOLDER QUESTION

Which assets have more confirmed failures than comparable assets under similar duty and exposure?

SENSORS AND TELEMETRY

Proposed evidence: fault, trip, unavailable-state, reset and runtime signals linked to stable asset identities. Coverage: comparable assets with consistent fault definitions and known duty; exclude commissioning and planned-test events. Timing: event capture with monthly and rolling-12-month summaries; aggregate per asset, peer group and failure mode. Quality limits: different control logic, alarm settings, duty cycles and recording practices bias comparisons. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: failure-coded work orders and asset class data, maintained by the Estates maintenance and asset information owners. Corroboration: BMS events, operator logs, service interruptions, commissioning records and replacement history. Precedence: validated work failures govern failure counts; alarms are corroborating evidence only. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag generic cause codes, duplicate work orders, parent-level booking, incomplete downtime and changed asset duty. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: calculation and comparison. Decision cadence: peer-normalised reliability benchmarking, monthly and quarterly for strategy decisions. Minimum evidence: at least three credible failures or a sufficient exposure period, peer definition, runtime or demand denominator and stable coding. Method: failure rate and recurrent-event analysis adjusted for exposure, with robust outlier detection. Uncertainty: rate intervals, peer-size limits and coding-quality score. Safe failure: report insufficient peer evidence rather than label an asset unreliable.

ANSWER BOUNDARY

Do not infer contractor competence, guarantee future reliability, or compare assets whose duty and coding are not materially comparable. The accountable maintenance owner retains authority over task, interval, closure and intervention decisions.

WHY THIS IS USEFUL

Reliability engineers can select credible peer outliers for investigation rather than ranking unlike assets.

EA-023

STAKEHOLDER QUESTION

Which failure symptoms recur after intervention, and where is competent root-cause work needed instead of another symptom-led response?

SENSORS AND TELEMETRY

Proposed evidence: fault-state, performance, alarm and reset sequences before and after work, using asset-specific signals. Coverage: assets with timestamped interventions and reliable state data. Timing: event-level capture with a defined post-work observation window; aggregate by asset, symptom and intervention type. Quality limits: normal commissioning checks, seasonal recurrence and incomplete fault clearance can mimic re-failure. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: work-order action, cause and completion evidence, maintained by the Estates maintenance owner. Corroboration: BMS trends, contractor reports, replaced-parts records, permits and operator notes. Precedence: validated work evidence determines the intervention; telemetry determines observed recurrence, not root cause. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag vague action codes, missing completion time, bundled assets and work closed before testing. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: bounded diagnosis. Decision cadence: pre/post intervention recurrence analysis, weekly screening with monthly engineering discussion. Minimum evidence: linked asset, credible intervention time, defined symptom, sufficient pre/post observation and comparable operating state. Method: temporal sequence matching with exclusion windows for testing and commissioning. Uncertainty: recurrence probability, time to return and evidence completeness. Safe failure: show a possible recurrence and request engineering diagnosis when cause or intervention scope is unclear.

ANSWER BOUNDARY

Do not diagnose root cause, declare work defective, assign blame or reopen a work order automatically. The accountable maintenance owner retains authority over task, interval, closure and intervention decisions.

WHY THIS IS USEFUL

Maintenance leads can distinguish symptom recurrence from a resolved outcome and request competent root-cause work.

EA-024

STAKEHOLDER QUESTION

Was planned maintenance completed on the correct asset, within the required window and with enough evidence to rely on the completion status?

SENSORS AND TELEMETRY

Proposed evidence: tagged attendance confirmation, approved test-result capture and post-task state observations where suitable. Coverage: planned tasks and assets in the authorised maintenance scope; no individual worker tracking. Timing: at task completion with daily schedule reconciliation; aggregate by task, asset, due window and evidence class. Quality limits: presence near an asset does not prove task completion, and a healthy state does not prove every task step occurred. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the authoritative maintenance schedule and work-order completion record, maintained by the Estates maintenance delivery owner. Corroboration: digital task sheets, test results, permits, contractor evidence, parts used and exceptions. Precedence: signed completion by an authorised role governs status; evidence sufficiency is assessed against the approved task requirement. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag late synchronisation, copied notes, missing test attachments, wrong asset links and retrospective closure. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: lookup and comparison. Decision cadence: schedule compliance and completion-evidence validation, daily with monthly service reporting. Minimum evidence: due date, task version, authorised completion, linked asset and all mandatory evidence fields. Method: date-window checks and evidence checklist validation, separating completed, completed-late, deferred and unverified. Uncertainty: status confidence and exact missing evidence. Safe failure: keep the record unverified and route to the maintenance owner when mandatory evidence is absent or contradictory.

ANSWER BOUNDARY

Do not certify task quality, identify individual performance, authorise closure or infer completion from attendance or telemetry alone. The accountable maintenance owner retains authority over task, interval, closure and intervention decisions.

WHY THIS IS USEFUL

Assurance reporting can separate verified completion from attendance, telemetry or unsupported closure.

EA-025

STAKEHOLDER QUESTION

Which non-mandatory tasks have enough sensing, failure-mode knowledge and fallback control to be considered for condition-based triggering?

SENSORS AND TELEMETRY

Proposed evidence: validated condition indicators, runtime, starts, load, environmental exposure and sensor-health data matched to the failure mode. Coverage: non-mandatory tasks on assets with sufficient sensing, history and safe fallback. Timing: continuous or periodic capture with quarterly strategy assessment; aggregate by task, asset class and failure mode. Quality limits: a measurable signal may not provide enough warning time, and sensing can miss latent failures. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: approved task library and maintenance strategy constraints, maintained by the Estates maintenance strategy owner. Corroboration: manufacturer guidance, warranty terms, statutory requirements, failure history, consequence analysis and sensor maintenance records. Precedence: mandatory obligations and competent engineering approval take precedence; analytics can only identify candidates. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag short histories, unrecorded failures, changed equipment, weak sensor coverage and tasks serving multiple failure modes. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: recommendation. Decision cadence: condition-based maintenance suitability assessment, quarterly and before strategy change. Minimum evidence: defined failure mode, measurable precursor, adequate lead time, healthy sensing, failure history and safe time-based fallback. Method: feasibility scoring plus retrospective simulation of trigger sensitivity, missed events and workload. Uncertainty: candidate confidence, false-trigger range and residual risk. Safe failure: recommend retaining the existing schedule when warning performance or evidence is inadequate.

ANSWER BOUNDARY

Do not change tasks, waive obligations, promise savings or claim that condition monitoring covers unobservable failure modes. The accountable maintenance owner retains authority over task, interval, closure and intervention decisions.

WHY THIS IS USEFUL

Owners can choose a bounded pilot with explicit fallback rather than reducing intervals estate-wide.

EA-026

STAKEHOLDER QUESTION

Which maintenance visits are delayed, repeated or abandoned because job information, parts, tools, permits or access arrangements are not ready?

SENSORS AND TELEMETRY

Proposed evidence: time-stamped asset scans, store issue events, delivery status and access-control status at an aggregated work-order level. Coverage: maintenance logistics processes, without tracking individual movement or revealing restricted access details. Timing: per work event with monthly pattern summaries; aggregate by asset class, missing prerequisite and service provider. Quality limits: late data entry, informal stock use and access denials for unrelated reasons may distort attribution. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: work-order delay and return-visit reasons, maintained by the Estates maintenance delivery owner. Corroboration: approved O&M instructions, parts catalogues, stock records, purchase orders, permits, access arrangements and contractor reports. Precedence: authorised work records and store or access owners confirm the constraint; telemetry only timestamps related events. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag generic delay codes, missing links to parts, outdated manuals, superseded tool requirements and hidden contractual boundaries. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: comparison and bounded diagnosis. Decision cadence: maintenance-friction and repeat-visit analysis, monthly and before contract or stock planning. Minimum evidence: linked work order, asset, delay or repeat attendance, coded prerequisite and corroborating transaction or document evidence. Method: process mining and Pareto analysis, separating controllable from external causes. Uncertainty: frequency, delay range and attribution confidence. Safe failure: report an unclassified constraint where evidence cannot distinguish parts, access, instruction or scheduling causes.

ANSWER BOUNDARY

Do not attribute delays to named individuals, expose access-control logs, order parts or alter contracts. The accountable maintenance owner retains authority over task, interval, closure and intervention decisions.

WHY THIS IS USEFUL

Planners can improve job packs, stores, permits and access readiness to reduce avoidable repeat visits.

EA-027

STAKEHOLDER QUESTION

Which assets or services take longest to restore after confirmed failure, and which restoration stages account for the delay?

SENSORS AND TELEMETRY

Proposed evidence: fault start, unavailable state, temporary restoration and stable service-state signals. Coverage: assets with reliable event timestamps and service definitions. Timing: event-level capture with monthly summaries; aggregate by asset, failure class and restoration stage. Quality limits: alarm time may differ from actual loss of service, and intermittent recovery can confuse restoration time. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: incident and work-order timelines, maintained by the Estates maintenance and service owners. Corroboration: call logs, attendance, diagnosis, parts procurement, access, isolation, testing, handback and service-owner confirmation. Precedence: service-owner confirmation governs restoration; system events provide observed state transitions. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag missing stage timestamps, parallel work orders, partial service restoration and inconsistent closure definitions. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: calculation and comparison. Decision cadence: mean and percentile time-to-restore decomposition, monthly and after major incidents. Minimum evidence: credible failure start, restoration definition, stage timestamps and at least several comparable events or clearly labelled single-event analysis. Method: survival and process-stage analysis, excluding planned outages and separating waiting from active work. Uncertainty: median, range, censored events and data-completeness limits. Safe failure: avoid comparative ranking and return a timeline gap when event definitions or timestamps are not comparable.

ANSWER BOUNDARY

Do not promise restoration times, judge individual performance, infer cause from duration or disclose sensitive incident details outside authorised roles. The accountable maintenance owner retains authority over task, interval, closure and intervention decisions.

WHY THIS IS USEFUL

Service owners can target the restoration stage that contributes most to confirmed downtime.

EA-028

STAKEHOLDER QUESTION

Which closed work orders lack enough asset, test or outcome evidence for estates teams to rely on the stated result?

SENSORS AND TELEMETRY

Proposed evidence: post-work asset state, test values and alarm clearance from commissioned sources, captured only where relevant to the task. Coverage: closed work orders on assets with defined completion evidence. Timing: at closure and after the approved proving period; aggregate by task type and missing evidence category. Quality limits: a normal state can be temporary, and many tasks cannot be verified remotely. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the closed work order and mandatory evidence schema, maintained by the Estates maintenance delivery owner. Corroboration: task sheets, test certificates, photographs under controlled access, parts records, commissioning results and service-owner handback. Precedence: authorised completion remains the record status, but evidence sufficiency follows the approved task standard. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag missing attachments, wrong asset links, generic outcome codes, contradictory test values and records created after closure. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: lookup and comparison. Decision cadence: closure-evidence sufficiency checking, daily and monthly for quality management. Minimum evidence: task version, asset link, authorised sign-off and all required evidence for that task. Method: schema validation plus consistency checks between action, parts, test and post-work state. Uncertainty: verified, incomplete or contradictory with exact reasons. Safe failure: flag for owner assessment; do not reopen or reject the work order automatically.

ANSWER BOUNDARY

Do not certify workmanship, expose restricted images, infer falsification or override an authorised closure. The accountable maintenance owner retains authority over task, interval, closure and intervention decisions.

WHY THIS IS USEFUL

The maintenance owner can request missing test or outcome evidence before relying on the work history.

EA-029

STAKEHOLDER QUESTION

Which maintenance windows satisfy isolation, access, resource and service constraints while causing the least credible disruption?

SENSORS AND TELEMETRY

Proposed evidence: aggregate room or zone utilisation, service demand, equipment duty and current environmental state. Coverage: affected spaces and systems, using privacy-preserving counts and no individual schedules. Timing: recent representative patterns with a forward window of days or weeks; aggregate by zone and time block. Quality limits: future events, research activity, access changes and unusual weather can invalidate historical patterns. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: approved maintenance deadline, access rules and service calendars, maintained by the Estates maintenance planner with space and service owners. Corroboration: teaching timetables, bookings, event schedules, permits, contractor availability, isolation requirements and risk controls. Precedence: current authorised schedules, permits and risk constraints take precedence over predicted low use. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag late bookings, restricted activities, stale occupancy models, unconfirmed resource availability and conflicting deadlines. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: forecast and recommendation. Decision cadence: constraint-based maintenance-window ranking, daily for the defined planning horizon. Minimum evidence: task duration, latest safe date, affected services, authorised access, resources, isolation needs and current schedule data. Method: feasible-window filtering followed by a transparent disruption score with scenario sensitivity. Uncertainty: rank range, excluded constraints and confidence in demand prediction. Safe failure: return no feasible window and escalate to planners when mandatory constraints conflict or evidence is stale.

ANSWER BOUNDARY

Do not book access, authorise isolation, guarantee low disruption, expose individual timetables or recommend delaying beyond a safe deadline. The accountable maintenance owner retains authority over task, interval, closure and intervention decisions.

WHY THIS IS USEFUL

Schedulers can identify feasible windows for accountable approval and stakeholder coordination.

EA-030

STAKEHOLDER QUESTION

Which reliability evidence supports considering a maintenance-strategy change, and which patterns justify only further investigation?

SENSORS AND TELEMETRY

Proposed evidence: validated failure, degradation, runtime, load and post-maintenance performance signals. Coverage: assets with sufficient history and stable identity; uninstrumented modes remain explicit gaps. Timing: monthly trend generation with quarterly strategy interpretation; aggregate by asset class, failure mode and duty. Quality limits: changes in coding, sensors, operating regime or fleet age can create artificial trends. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: approved reliability measures and maintenance history, maintained by the Estates maintenance strategy owner. Corroboration: asset changes, work orders, service interruptions, condition surveys, spares use and project interventions. Precedence: competent engineering decisions govern strategy changes; analytics supplies evidence and counter-evidence. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag small samples, censored history, merged assets, changed failure definitions and missing exposure data. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: comparison and recommendation. Decision cadence: reliability trend classification, quarterly and before strategy approval. Minimum evidence: stable definitions, adequate exposure, credible failure events, known interventions and comparable operating periods. Method: trend and change-point tests, peer comparison and sensitivity to coding or duty changes. Uncertainty: effect size, confidence interval, sample sufficiency and plausible confounders. Safe failure: classify as inconclusive and specify the next evidence needed when the trend is not robust.

ANSWER BOUNDARY

Do not alter strategy, guarantee enhancement, infer causation from correlation or conceal weak samples behind a single trend line. The accountable maintenance owner retains authority over task, interval, closure and intervention decisions.

WHY THIS IS USEFUL

Owners can decide whether evidence supports a controlled strategy trial or only further investigation.

EA-031

STAKEHOLDER QUESTION

Who is currently accountable for each space, asset and shared service, and where are ownership, operation or funding responsibilities unclear?

SENSORS AND TELEMETRY

Proposed evidence: none as authority; proposed asset scans and service-topology evidence can expose where operational control appears inconsistent with recorded responsibility. Coverage: spaces, assets and shared services in the approved governance scope. Timing: monthly exception refresh and after organisational or project change; aggregate by responsibility type, not by named individual in general reports. Quality limits: operational use or control signals cannot establish legal, budget or safety accountability. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the approved responsibility and ownership matrix, maintained by the Estates governance owner with school, service and finance owners. Corroboration: space allocations, maintenance contracts, lease or licence records, asset ownership fields, budget codes and service agreements. Precedence: signed governance, tenure and contract records take precedence; observed operation only prompts clarification. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag vacant roles, expired agreements, inherited budget codes, conflicting owner fields and shared systems without a lead owner. Preserve unresolved conflicts for the accountable owner; never select a convenient value. The May 2021 floor plan is historical reference only and cannot establish current use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation: lookup and comparison. Decision cadence: responsibility-gap and conflict detection, monthly and at change gates. Minimum evidence: asset or space identity, responsibility type, current owner role, effective dates and supporting agreement. Method: matrix completeness checks and conflict rules across operational, maintenance, budget and assurance responsibilities. Uncertainty: confirmed, conflicting or unassigned responsibility with source dates. Safe failure: return an ownership question to authorised governance roles when no current record resolves it.

ANSWER BOUNDARY

Do not assign accountability, infer it from who uses or controls an asset, disclose personal details or interpret contractual liability. Approved space, tenure, access and information-governance records retain authority over use and disclosure.

WHY THIS IS USEFUL

Governance leads can resolve responsibility gaps before costs, risks or work are assigned.

EA-032

STAKEHOLDER QUESTION

Does recorded departmental tenure or authorised space allocation match the current approved arrangement and observed operational context?

SENSORS AND TELEMETRY

Proposed evidence: privacy-preserving aggregate entry counts, environmental utilisation indicators and room booking status. Coverage: spaces where aggregate use evidence is permitted; no identity, device tracking or research-content inspection. Timing: 15-minute or hourly aggregation, assessed over representative teaching and non-teaching periods; aggregate by room or zone with suppression for low counts. Quality limits: presence does not establish tenure, and low observed use may be legitimate or seasonal. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the current authorised space allocation, lease or licence record, maintained by the Estates Space Management owner. Corroboration: approved room directory, departmental confirmation, booking rules, project decants and access arrangements. Precedence: signed allocation and current accountable owner confirmation take precedence over observed use. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag temporary occupation, delayed directory updates, shared spaces, closed areas and conflicting effective dates. Preserve unresolved conflicts for the accountable owner; never select a convenient value. The May 2021 floor plan is historical reference only and cannot establish current use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation: observation and comparison. Decision cadence: allocation-versus-use exception detection, monthly and after known moves. Minimum evidence: current allocation, effective dates, representative aggregate use and known temporary arrangements. Method: rule comparison with seasonal baselines and explicit categories for shared, temporarily occupied and inactive. Uncertainty: exception confidence and observation coverage. Safe failure: state that use evidence is inconclusive and request owner confirmation when patterns are sparse or legitimate alternatives exist.

ANSWER BOUNDARY

Do not identify occupants, infer unauthorised occupation, change allocation, disclose sensitive room use or treat utilisation as ownership. Approved space, tenure, access and information-governance records retain authority over use and disclosure.

WHY THIS IS USEFUL

Space managers can verify an apparent mismatch with the authorised allocation owner rather than treating presence as tenure.

EA-033

STAKEHOLDER QUESTION

Which spaces show persistent aggregate underuse, crowding or service-intensity patterns that merit a space-management check?

SENSORS AND TELEMETRY

Proposed evidence: non-identifying people counts, doorway counts, CO2 as a secondary indicator, booking check-in state and environmental occupancy proxies. Coverage: approved rooms or zones; suppress small counts and exclude spaces where privacy or research sensitivity prevents monitoring. Timing: 5-or 15-minute sensing aggregated to hourly, daily and representative-week patterns; aggregate by space type and usable time, not by individual. Quality limits: sensor occlusion, open doors, shared ventilation, unrecorded events and inaccessible areas affect counts. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: current room capacity, function and availability, maintained by the Estates Space Management owner. Corroboration: timetables, bookings, closures, accessibility constraints, service schedules and approved local use rules. Precedence: current authorised capacity, closures and function take precedence over historical plans or sensor estimates. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag changed room layouts, cancelled bookings, temporary restrictions, stale capacity and conflicting room names. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: observation and comparison. Decision cadence: aggregate utilisation and capacity-pressure analysis, weekly, termly and for specified planning periods. Minimum evidence: verified room identity, current usable capacity, sufficient monitored hours, booking context and closure data. Method: occupancy-to-capacity distributions with time-of-day and term controls; no productivity interpretation. Uncertainty: coverage, count error range and persistence across periods. Safe failure: return no conclusion where monitoring coverage, current function or capacity is unverified.

ANSWER BOUNDARY

Do not identify users, infer attendance or performance, declare a space surplus, authorise reallocations or use the May 2021 plan as proof of current function. Approved space, tenure, access and information-governance records retain authority over use and disclosure.

WHY THIS IS USEFUL

Space teams can select areas for a management check without identifying individuals or inferring productivity.

EA-034

STAKEHOLDER QUESTION

Which assets and services support several departments or tenants, and what responsibility or cost-sharing basis is currently authorised?

SENSORS AND TELEMETRY

Proposed evidence: system-level flow, energy, runtime and service-state metering where commissioned and technically attributable. Coverage: shared systems and served zones at a level that does not reveal sensitive activity. Timing: hourly or daily aggregation with monthly allocation summaries; aggregate by service and authorised responsibility unit. Quality limits: submeter gaps, common losses, shared plant and changing use prevent exact attribution. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: service responsibility agreements and cost-allocation rules, maintained by the Estates governance and finance owners. Corroboration: asset-to-space topology, space allocations, leases or licences, metering maps, maintenance contracts and service-level records. Precedence: signed agreements and finance policy take precedence over measured demand; telemetry may inform but not redefine liability. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag unmetered branches, changed tenants, temporary decants, missing shared-cost rules and topology conflicts. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: calculation and estimate. Decision cadence: shared-service mapping and bounded allocation support, monthly and at agreement change. Minimum evidence: verified served-area map, current occupants, approved allocation rule and sufficient metering or declared proxy. Method: dependency graph plus rule-based allocation with transparent residual or common-service share. Uncertainty: attribution range and unallocated proportion. Safe failure: show an unallocatable shared portion rather than force exact splits when topology or metering is inadequate.

ANSWER BOUNDARY

Do not create charges, interpret leases, expose one tenant's sensitive demand or assign legal responsibility. Approved space, tenure, access and information-governance records retain authority over use and disclosure.

WHY THIS IS USEFUL

Accountable owners can clarify service, maintenance and cost-sharing arrangements for shared infrastructure.

EA-035

STAKEHOLDER QUESTION

Which spaces have operational constraints that are missing, stale or inconsistent across estate information?

SENSORS AND TELEMETRY

Proposed evidence: door or plant state, environmental limits, access-status summaries and equipment-operation windows where already approved. Coverage: space-level constraints such as restricted access, environmental dependence or service shutdown windows; no personal monitoring. Timing: event-driven with monthly reconciliation; aggregate as controlled constraint categories, with detail restricted by role. Quality limits: a closed door, unusual temperature or low use does not prove a formal constraint. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the current space constraint and operational information record, maintained by the Estates Space Management owner with safety, security and local service owners. Corroboration: approved access rules, hazard records, business continuity needs, room data sheets, project restrictions and local operating agreements. Precedence: current authorised safety, security and service-owner records take precedence; the May 2021 plan has no authority for current constraints. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag temporary restrictions without end dates, confidential constraints, inconsistent room identifiers and unrecorded local arrangements. Preserve unresolved conflicts for the accountable owner; never select a convenient value. The May 2021 floor plan is historical reference only and cannot establish current use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation: lookup and comparison. Decision cadence: constraint completeness and exception detection, monthly and before planning work or access. Minimum evidence: current space identity, recorded constraints, an authorised source or repeated operational exception and effective dates. Method: rule checks across access, environmental, service and hazard categories followed by owner confirmation. Uncertainty: confirmed, probable or unverified constraint with disclosure level. Safe failure: show only that confirmation is required when details are restricted or evidence is indirect.

ANSWER BOUNDARY

Do not infer hazards, grant access, reveal restricted activities, declare a room safe or instruct entry. Approved space, tenure, access and information-governance records retain authority over use and disclosure.

WHY THIS IS USEFUL

Operational constraints can be corrected before they invalidate access, project or maintenance planning.

EA-036

STAKEHOLDER QUESTION

Where are room names, numbers, functions or boundaries inconsistent across space, asset, maintenance and project systems?

SENSORS AND TELEMETRY

Proposed evidence: tagged room-sign observations and permissioned survey geometry where needed. Coverage: spaces within the agreed information scope; no capture of occupants or room content. Timing: after layout or signage change and quarterly exception sampling; aggregate by canonical space identifier and conflicting field. Quality limits: temporary signs, inaccessible rooms and survey-coordinate error can create false conflicts. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the current canonical space hierarchy, maintained by the Estates Space Information owner. Corroboration: approved plans, room directory, timetabling, access control, CAFM, BIM or CDE and signage records. Precedence: the owner-approved current space identifier and boundary govern; downstream systems must not be reconciled by majority vote. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag renumbering, subdivisions, merged rooms, delayed interfaces, historical plan references and conflicting effective dates. Preserve unresolved conflicts for the accountable owner; never select a convenient value. The May 2021 floor plan is historical reference only and cannot establish current use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation: comparison. Decision cadence: cross-system space identity reconciliation, weekly for changed records and quarterly across the estate. Minimum evidence: canonical identifier, current boundary or parent, source timestamps and at least one approved corroborating record. Method: exact-key checks, alias mapping and geometry overlap only where current approved geometry exists. Uncertainty: conflict type, affected systems and match confidence. Safe failure: retain separate candidates and request space-owner decision where current boundary or identifier is not authoritative.

ANSWER BOUNDARY

Do not rename rooms, update downstream systems, reveal restricted space functions or use the May 2021 plan as current authority. Approved space, tenure, access and information-governance records retain authority over use and disclosure.

WHY THIS IS USEFUL

Information owners can reconcile identifiers and boundaries so records and work refer to the same current space.

EA-037

STAKEHOLDER QUESTION

Which spaces are currently unavailable or partly available, and what authoritative reason, period and reopening condition are recorded?

SENSORS AND TELEMETRY

Proposed evidence: approved access-state, environmental alarms, service availability and non-identifying aggregate use indicators. Coverage: spaces with an authorised availability status; restricted reasons remain role-limited. Timing: near real time for state, with daily reconciliation to formal records; aggregate by space and availability category. Quality limits: access-control state, low use or environmental alarm alone cannot establish closure. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the authorised closure or availability record, maintained by the Estates operations or Space Management owner. Corroboration: project restrictions, incidents, maintenance work, safety decisions, bookings and service-owner notices. Precedence: current accountable owner status and competent safety decisions take precedence over sensors, bookings and historical plans. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag closures without end conditions, expired notices, partial restrictions, conflicting access state and delayed reopening updates. Preserve unresolved conflicts for the accountable owner; never select a convenient value. The May 2021 floor plan is historical reference only and cannot establish current use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation: lookup and observation. Decision cadence: availability-state reconciliation, near real time with daily owner confirmation for prolonged exceptions. Minimum evidence: current authorised status, effective time, reason category, affected scope, end condition and any corroborating service state. Method: state-machine checks and conflict detection across closure, booking, access and service records. Uncertainty: confirmed, conflicting or unknown availability. Safe failure: display unknown and direct the user to the current owner when no authoritative state exists.

ANSWER BOUNDARY

Do not declare a space safe, open or closed, grant entry, reveal sensitive incident reasons or promise a reopening time. Approved space, tenure, access and information-governance records retain authority over use and disclosure.

WHY THIS IS USEFUL

Service teams can communicate a permissioned availability status and the condition for a governed reopening decision.

EA-038

STAKEHOLDER QUESTION

How could a proposed change of space use affect the assets, capacities, controls and shared services supporting the area?

SENSORS AND TELEMETRY

Proposed evidence: current aggregate demand, environmental performance, service load and equipment duty relevant to the proposed change. Coverage: affected rooms, zones and upstream plant; privacy-preserving aggregation only. Timing: representative recent periods and scenario-specific design windows; aggregate by service and capacity margin. Quality limits: historical demand may not represent the proposed activity, and hidden routes or unmetered loads remain uncertain. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the current approved space and system topology, maintained by the Estates Space Information and engineering owners. Corroboration: project brief, proposed occupancy or activity, design criteria, asset capacities, O&M limits, statutory constraints and current defects. Precedence: approved current records and competent design input take precedence; the historical May 2021 plan may only orient, not establish present conditions. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag unverified capacities, undocumented changes, unresolved defects, incomplete project assumptions and conflicting room boundaries. Preserve unresolved conflicts for the accountable owner; never select a convenient value. The May 2021 floor plan is historical reference only and cannot establish current use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation: estimate and forecast. Decision cadence: change-impact and capacity-screening scenario, at concept, option and design decision points. Minimum evidence: defined proposed use, current verified space and service relationships, representative demand and capacity or constraint evidence. Method: dependency tracing and bounded peak-demand comparison under low, central and high scenarios. Uncertainty: capacity margin ranges and named evidence gaps. Safe failure: return a requirement for survey or design confirmation when current topology, capacity or future demand is not adequate.

ANSWER BOUNDARY

Do not approve the change, certify capacity, infer design compliance, authorise works or replace professional design and survey activity. Approved space, tenure, access and information-governance records retain authority over use and disclosure.

WHY THIS IS USEFUL

Project teams can identify enabling studies and service constraints before a proposed use change advances.

EA-039

STAKEHOLDER QUESTION

Which assets cannot be maintained reliably because access routes, permissions, keys, permits or working windows are absent or repeatedly ineffective?

SENSORS AND TELEMETRY

Proposed evidence: aggregated access-denial events, tagged attendance at the asset zone and door-state summaries where permitted. Coverage: maintenance access paths and plant areas; no personal movement profiles or disclosure of security details. Timing: per work event with monthly pattern analysis; aggregate by asset, access constraint type and responsible access owner. Quality limits: access denial can reflect expired credentials, timing, escort absence or unrelated security events. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: approved asset access requirements and work-order delay reasons, maintained by the Estates maintenance and security or local access owners. Corroboration: permits, escort arrangements, keys, room restrictions, service windows and contractor reports. Precedence: current access and safety owners decide permission; repeated operational evidence only identifies a gap. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag local arrangements not recorded, changed locks, temporary restrictions, generic delay codes and conflicting access instructions. Preserve unresolved conflicts for the accountable owner; never select a convenient value. The May 2021 floor plan is historical reference only and cannot establish current use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation: comparison and bounded diagnosis. Decision cadence: access-readiness and repeat-failure analysis, monthly and before planned high-priority work. Minimum evidence: asset and safe access requirement, at least one failed or delayed attendance, authorised access record and current owner. Method: process-stage analysis separating permission, credential, key, escort, timing and hazard prerequisites. Uncertainty: constraint category and attribution confidence. Safe failure: state that access readiness is unverified and route to the current owner when event detail is insufficient or restricted.

ANSWER BOUNDARY

Do not grant access, reveal credentials or sensitive routes, identify individual movements, or direct entry without current permission and safe-system controls. Approved space, tenure, access and information-governance records retain authority over use and disclosure.

WHY THIS IS USEFUL

Maintenance planners can resolve permits, keys, routes or working-window dependencies before booking work.

EA-040

STAKEHOLDER QUESTION

What asset and space information may be shown to this user role for this purpose without exposing restricted or personal detail?

SENSORS AND TELEMETRY

Proposed evidence: none as authority; data classifications from sensor metadata can indicate whether a feed contains location, security or low-count occupancy information. Coverage: all proposed outputs and source fields. Timing: at design time and whenever source, purpose or audience changes; aggregate by role, purpose and minimum necessary field set. Quality limits: technical metadata cannot determine lawful or institutional permission by itself. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the approved information-classification and access-control policy, maintained by the Estates information governance and security owners. Corroboration: data-owner permissions, contractual controls, research restrictions, privacy assessments and system role mappings. Precedence: current policy and explicit data-owner permission take precedence over convenience or inherited access. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag role drift, copied exports, stale group membership, unclear purpose and conflicts between source permissions. Preserve unresolved conflicts for the accountable owner; never select a convenient value. The May 2021 floor plan is historical reference only and cannot establish current use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation: lookup. Decision cadence: purpose-based disclosure filtering, at every query and permission change. Minimum evidence: authenticated role, approved purpose, source permission, field classification, location sensitivity and aggregation rule. Method: deny-by-default policy evaluation with field-level minimisation, low-count suppression and output logging. Uncertainty: allowed fields plus any redaction or aggregation applied. Safe failure: refuse the restricted portion and return the responsible data owner or safer aggregate where available.

ANSWER BOUNDARY

Do not broaden access, reveal restricted plant or research information, expose individual-level occupancy, or assume that access to one source permits redistribution. Approved space, tenure, access and information-governance records retain authority over use and disclosure.

WHY THIS IS USEFUL

The service can release the minimum necessary asset or space detail for the current role and purpose.

EA-041

STAKEHOLDER QUESTION

Which planned projects, maintenance packages or operational changes affect the same asset, space or service, and where could scopes or timings clash?

SENSORS AND TELEMETRY

Proposed evidence: current asset and service state plus tagged verification of temporary works or active project zones where approved. Coverage: assets, spaces and dependencies in the selected planning horizon. Timing: daily project-feed refresh and event updates for material state change; aggregate by asset, space, service and time window. Quality limits: site activity and temporary states may not be captured consistently, and telemetry cannot define contractual scope. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the authorised project portfolio, scope and programme records, maintained by the Estates Capital Projects owner. Corroboration: change requests, work packages, outage plans, current asset hierarchy, space closures, maintenance plans and local service constraints. Precedence: approved project and operational owner records take precedence; inferred overlap only raises a coordination question. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag uncertain dates, unapproved options, different space identifiers, omitted enabling works and conflicting programme versions. Preserve unresolved conflicts for the accountable owner; never select a convenient value. The May 2021 floor plan is historical reference only and cannot establish current use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation: comparison and forecast. Decision cadence: project-to-asset dependency and schedule-clash detection, daily for the next 12 weeks and at project gates. Minimum evidence: current project scope, affected identities, dates, service dependencies and operational constraints. Method: graph intersection plus interval-overlap and prerequisite rules. Uncertainty: confirmed and possible clashes with source-date confidence. Safe failure: show an unresolved coordination risk and request owner confirmation when scope or dates are provisional.

ANSWER BOUNDARY

Do not approve scope, change programme dates, expose commercially restricted details, authorise outages or declare conflicts resolved. Approved project governance and accountable operational owners retain authority over scope, acceptance and handover.

WHY THIS IS USEFUL

Programme teams can coordinate interfaces, shutdowns and ownership before overlapping scopes create rework or service risk.

EA-042

STAKEHOLDER QUESTION

What baseline evidence must be captured before work starts so condition, performance, energy and configuration changes can later be demonstrated?

SENSORS AND TELEMETRY

Proposed evidence: asset-specific baseline trends, meter readings, configuration exports, tag and serial verification, and controlled condition observations. Coverage: assets, services and spaces within the approved project impact boundary. Timing: for a representative pre-work period plus a timestamped final baseline close to commencement; aggregate by asset, service level and relevant operating condition. Quality limits: seasonal variability, temporary operation, poor sensor health and short baselines can make comparison misleading. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the approved project information requirements and current asset or space records, maintained by the Estates Capital Projects and Information Management owners. Corroboration: design intent, current O&M data, condition surveys, service criteria, risk records and known defects. Precedence: the agreed information requirement and competent survey scope determine the baseline; telemetry supplements but does not replace them. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag late scope changes, missing current drawings, unresolved defects, uncalibrated sensors and uncertain operating context. Preserve unresolved conflicts for the accountable owner; never select a convenient value. The May 2021 floor plan is historical reference only and cannot establish current use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation: observation and baseline calculation. Decision cadence: baseline sufficiency design, at project definition and before commencement. Minimum evidence: defined change claim, affected scope, current authoritative records, representative operating states and measurement quality. Method: claim-to-evidence mapping with coverage, duration and comparability checks. Uncertainty: which claims can be tested and the residual uncertainty. Safe failure: state that no reliable before-and-after comparison will be possible for unsupported claims and escalate the capture gap.

ANSWER BOUNDARY

Do not certify existing condition, approve project start, require intrusive measurement without permission or capture sensitive activity beyond the stated purpose. Approved project governance and accountable operational owners retain authority over scope, acceptance and handover.

WHY THIS IS USEFUL

Project and operational owners can agree a traceable baseline capable of supporting later comparison.

EA-043

STAKEHOLDER QUESTION

Have completed changes altered an asset's identity, location, duty, capacity, controls or dependencies in ways operational records must reflect?

SENSORS AND TELEMETRY

Proposed evidence: post-change tag scans, controller mappings, operating points, capacity test results and service-state observations. Coverage: changed assets and directly affected upstream or downstream systems. Timing: at commissioning, handover and after the proving period; aggregate as before-and-after attribute and relationship differences. Quality limits: temporary commissioning settings and incomplete proving can look like final duty. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the approved change record and accepted handover information, maintained by the Estates project and Asset Information owners. Corroboration: as-built documents, commissioning results, work orders, configuration records, approved substitutions and disposal evidence. Precedence: accepted change and accountable owner confirmation take precedence; observed differences create exceptions until accepted. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag red-line drawings not incorporated, undocumented substitutions, temporary settings, missing predecessor links and conflicting acceptance dates. Preserve unresolved conflicts for the accountable owner; never select a convenient value. The May 2021 floor plan is historical reference only and cannot establish current use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation: comparison. Decision cadence: before-and-after change reconciliation, at handover and after the agreed proving period. Minimum evidence: pre-change baseline, approved scope, installed identity, accepted test evidence and current owner confirmation. Method: attribute diff, relationship graph comparison and state validation against design intent. Uncertainty: confirmed, provisional or unexplained change with source trace. Safe failure: retain both prior and proposed states and request acceptance evidence when the final state is not authoritative.

ANSWER BOUNDARY

Do not update asset records, certify capacity, approve substitutions, infer project acceptance or disclose restricted configuration. Approved project governance and accountable operational owners retain authority over scope, acceptance and handover.

WHY THIS IS USEFUL

Asset-information owners can target post-change record updates and verification before stale configuration is relied upon.

EA-044

STAKEHOLDER QUESTION

Has handover supplied the asset, O&M, commissioning, as-built and maintenance information required for safe operational use?

SENSORS AND TELEMETRY

Proposed evidence: commissioning results, point-to-asset mappings, trend availability and asset tag verification as evidence of implemented information. Coverage: handover assets and systems defined in the project information requirement. Timing: at staged handover, practical completion and end of proving; aggregate by required deliverable, asset and acceptance status. Quality limits: a live point or tagged asset does not prove that documentation is complete, correct or approved. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the approved exchange or handover information requirement, maintained by the Estates Information Management and Capital Projects owners. Corroboration: CDE submissions, asset data sheets, O&M manuals, commissioning records, training records, warranties and defect lists. Precedence: formally accepted deliverables and current accountable owner decisions take precedence over file presence. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag wrong versions, unapproved status, inaccessible links, incomplete asset coverage, invalid formats and conflicting serials. Preserve unresolved conflicts for the accountable owner; never select a convenient value. The May 2021 floor plan is historical reference only and cannot establish current use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation: lookup and comparison. Decision cadence: handover completeness and usability validation, at each handover gate. Minimum evidence: required-item matrix, approved status, asset coverage, accessible file, owner, version and effective date. Method: schema, coverage, status and link-integrity checks plus sample trace from asset to required documents. Uncertainty: complete, conditional or missing with exact acceptance dependencies. Safe failure: withhold a complete status and list the unresolved deliverables when acceptance or linkage is absent.

ANSWER BOUNDARY

Do not accept handover, certify information accuracy, approve payment, expose commercial documents or substitute for technical checking. Approved project governance and accountable operational owners retain authority over scope, acceptance and handover.

WHY THIS IS USEFUL

The receiving owner can identify missing handover evidence before accepting operational support responsibility.

EA-045

STAKEHOLDER QUESTION

Which handover records cannot be matched confidently to the installed assets they are intended to describe?

SENSORS AND TELEMETRY

Proposed evidence: tag, serial, make, model, controller object and location verification. Coverage: installed assets within the project boundary, excluding items outside the agreed estate asset scope. Timing: during installation checks, handover and first maintenance attendance; aggregate as unmatched records and candidate assets. Quality limits: serial plates may be inaccessible or replaced, and identical units can share descriptions. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the accepted project asset schedule, maintained by the Estates Information Management owner. Corroboration: asset register, procurement records, commissioning sheets, warranty registrations, photographs under controlled access and BIM objects. Precedence: physical identity confirmed by the accountable asset owner governs matching; a project row cannot create an asset alone. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag duplicate serials, generic descriptions, wrong locations, substitutions and missing predecessor or package relationships. Preserve unresolved conflicts for the accountable owner; never select a convenient value. The May 2021 floor plan is historical reference only and cannot establish current use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation: comparison and entity resolution. Decision cadence: handover entity matching, at each data drop and handover gate. Minimum evidence: at least two independent identifiers or one unique serial plus credible location or relationship evidence. Method: deterministic key matching followed by weighted similarity and one-to-many package checks. Uncertainty: match confidence and attributes causing ambiguity. Safe failure: leave the row unmatched and request site or supplier confirmation when identity cannot be resolved.

ANSWER BOUNDARY

Do not invent identifiers, merge installed assets, approve substitutions, expose restricted serials or treat similarity as acceptance. Approved project governance and accountable operational owners retain authority over scope, acceptance and handover.

WHY THIS IS USEFUL

Handover teams can target identity checks instead of attaching records to the wrong installed asset.

EA-046

STAKEHOLDER QUESTION

Are approved project changes represented consistently in the asset register, BIM or CDE, controlled drawings, BMS and maintenance systems?

SENSORS AND TELEMETRY

Proposed evidence: commissioned control-point inventories and tagged field observations that can confirm selected implemented changes. Coverage: changed assets and relationships named in the approved change set. Timing: after acceptance and then daily interface checks until reconciled; aggregate by change item and target system. Quality limits: not every change is observable, and live systems may contain temporary commissioning states. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the accepted change set and system-of-record map, maintained by the Estates Information Management owner with each application owner. Corroboration: asset register, BIM or CDE, drawings, BMS configuration, maintenance plans and document transmittals. Precedence: the named authoritative owner for each field governs; consistency does not mean every system is authoritative for every attribute. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag interface latency, superseded models, local copies, incomplete mappings and changes accepted at different dates. Preserve unresolved conflicts for the accountable owner; never select a convenient value. The May 2021 floor plan is historical reference only and cannot establish current use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation: comparison. Decision cadence: cross-system change propagation checking, daily after handover and monthly thereafter for unresolved items. Minimum evidence: approved change identifier, affected fields, target systems, expected effective date and provenance. Method: field-level reconciliation using authoritative-source mapping and allowed propagation delays. Uncertainty: consistent, pending, conflicting or not-applicable state. Safe failure: do not force consistency; show the authoritative value and route each conflict to the relevant system owner.

ANSWER BOUNDARY

Do not write to source systems, select a value by majority, expose restricted controls information or declare handover complete. Approved project governance and accountable operational owners retain authority over scope, acceptance and handover.

WHY THIS IS USEFUL

Information owners can reconcile approved change across operational systems before conflicting configurations affect work.

EA-047

STAKEHOLDER QUESTION

Which temporary assets, bypasses, overrides, supplies or controls may still be present after their approved work period ended?

SENSORS AND TELEMETRY

Proposed evidence: temporary-controller objects, override flags, bypass position, temporary meter or gateway heartbeats and tagged site verification. Coverage: the completed work boundary and linked systems, limited to approved controls data. Timing: near real time for flagged states and at completion or demobilisation checks; aggregate by temporary item, responsible work package and expiry condition. Quality limits: legitimate seasonal overrides, retained monitoring and stale control flags can resemble abandoned temporary arrangements. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the approved temporary-works and override register, maintained by the Estates project, controls and safety owners. Corroboration: permits, commissioning plans, change logs, completion records, asset schedules and defect lists. Precedence: current authorised temporary-control records and competent engineering decisions take precedence over automated detection. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag missing expiry dates, unclosed permits, undocumented retained devices, project-system closure before operational removal and conflicting owners. Preserve unresolved conflicts for the accountable owner; never select a convenient value. The May 2021 floor plan is historical reference only and cannot establish current use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation: observation and comparison. Decision cadence: temporary-state persistence detection, daily until all approved end conditions are confirmed. Minimum evidence: identified temporary item or state, authorised purpose, responsible owner, expected end condition and current corroborating evidence. Method: state-duration and expiry rules with checks against active permits and accepted retained scope. Uncertainty: confirmed, possible or unexplained persistence. Safe failure: raise an urgent owner query without instructing removal when purpose, safety state or authority is uncertain.

ANSWER BOUNDARY

Do not remove or disable devices, clear overrides, authorise normal operation, disclose sensitive control states or declare a system safe. Approved project governance and accountable operational owners retain authority over scope, acceptance and handover.

WHY THIS IS USEFUL

The responsible owner can verify and close out temporary arrangements that may otherwise become unmanaged dependencies.

EA-048

STAKEHOLDER QUESTION

Are project defects and outstanding works linked to the correct asset, responsible party, evidence, due condition and closeout authority?

SENSORS AND TELEMETRY

Proposed evidence: persistent fault or service-state evidence and tagged asset verification where relevant. Coverage: handover defects and outstanding items within the approved project scope. Timing: daily status refresh through the defect period; aggregate by defect, asset, responsibility and end condition. Quality limits: telemetry cannot establish contractual responsibility or prove cosmetic or hidden defects. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the authoritative defect and outstanding-work register, maintained by the Estates Capital Projects owner. Corroboration: contracts, accepted scope, work packages, asset data, inspection records, correspondence and completion evidence. Precedence: the authorised contract and project owner determine responsibility and closure; sensor evidence only supports current observed state. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag generic locations, disputed scope, changed due dates, duplicate defects, missing end conditions and conflicting acceptance records. Preserve unresolved conflicts for the accountable owner; never select a convenient value. The May 2021 floor plan is historical reference only and cannot establish current use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation: lookup and comparison. Decision cadence: defect linkage and responsibility completeness checking, daily and at project governance points. Minimum evidence: defect identifier, asset or space link, scope reference, responsible role, target or end condition and evidence status. Method: schema validation, entity matching and dependency checks between defect, work package and acceptance evidence. Uncertainty: complete, ambiguous or disputed linkage. Safe failure: preserve disputed status and route to authorised project or contract roles rather than resolving it analytically.

ANSWER BOUNDARY

Do not assign liability, certify completion, approve payment, expose privileged correspondence or close defects. Approved project governance and accountable operational owners retain authority over scope, acceptance and handover.

WHY THIS IS USEFUL

Project owners can route each outstanding item to the correct asset, party and closeout authority.

EA-049

STAKEHOLDER QUESTION

What operational-readiness evidence is still missing before estates teams can support the changed asset or service?

SENSORS AND TELEMETRY

Proposed evidence: stable performance during the proving period, alarm routing, control-point health, metering availability and tag verification. Coverage: changed assets and supporting systems in the handover boundary. Timing: continuous or scheduled proving with gate-based summaries; aggregate by readiness criterion and responsible owner. Quality limits: short stable operation does not demonstrate all seasons, failure modes or maintainability. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the approved operational-readiness criteria, maintained by the Estates operations and Capital Projects owners. Corroboration: commissioning, training, spares, maintenance plans, warranties, access arrangements, safety information, defects and service-owner acceptance. Precedence: formal acceptance and competent technical decisions take precedence; analytics cannot transfer responsibility. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag incomplete training evidence, conditional tests, open critical defects, missing support contracts and unverified alarm routes. Preserve unresolved conflicts for the accountable owner; never select a convenient value. The May 2021 floor plan is historical reference only and cannot establish current use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation: comparison and recommendation. Decision cadence: readiness-gate evidence assessment, at each acceptance gate and before support transfer. Minimum evidence: the criterion set, accepted test evidence, named operational owner, support prerequisites and unresolved defects. Method: binary and conditional gate checks with dependency ordering and explicit waivers only when formally recorded. Uncertainty: ready, conditionally ready or not evidenced with exact blockers. Safe failure: return not evidenced and list responsible confirmation routes when any mandatory criterion lacks current authority.

ANSWER BOUNDARY

Do not accept the asset, certify safe operation, authorise use, approve a waiver or promise support capacity. Approved project governance and accountable operational owners retain authority over scope, acceptance and handover.

WHY THIS IS USEFUL

Operational owners can identify the evidence, training, spares or controls still needed for support readiness.

EA-050

STAKEHOLDER QUESTION

What information must be retained, linked or access-restricted when an asset is replaced, moved or disposed of?

SENSORS AND TELEMETRY

Proposed evidence: final meter or runtime readings, configuration export, tag and serial verification, and decommissioned-state evidence where permitted. Coverage: the outgoing asset, successor and affected system relationships. Timing: immediately before change and after accepted completion; aggregate as a linked lifecycle package with access controls. Quality limits: final readings may be unavailable, and telemetry silence cannot prove physical disposal. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the records-retention and asset-disposal requirements, maintained by the Estates Information Governance, Finance and Asset Information owners. Corroboration: work history, condition, statutory evidence, warranties, configuration, drawings, finance records, environmental disposal evidence and project change records. Precedence: approved retention policy, finance requirements and accountable owner decisions govern; the successor record must not erase predecessor history. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag orphaned attachments, broken links, unrecorded successor relationships, conflicting disposal dates and restricted data mixed with general records. Preserve unresolved conflicts for the accountable owner; never select a convenient value. The May 2021 floor plan is historical reference only and cannot establish current use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation: lookup and recommendation. Decision cadence: lifecycle record closure and retention checking, at change approval, completion and scheduled retention points. Minimum evidence: outgoing and successor identities, change authority, required record classes, retention period, disposal evidence and access classification. Method: retention matrix validation plus predecessor-successor link and archive integrity checks. Uncertainty: complete, restricted, missing or not-yet-due status. Safe failure: prevent automated purge and escalate to record owners when requirements or legal holds are unclear.

ANSWER BOUNDARY

Do not authorise disposal, determine residual value, delete records, certify environmental treatment or expose restricted historical information. Approved project governance and accountable operational owners retain authority over scope, acceptance and handover.

WHY THIS IS USEFUL

Records teams can preserve lifecycle traceability while restricting sensitive, personal or commercially controlled content.

EA-051

STAKEHOLDER QUESTION

Which assets are within a recorded warranty period, and what conditions, evidence or notification deadlines could affect a potential claim?

SENSORS AND TELEMETRY

Proposed evidence: runtime, starts, environmental exposure, alarm history and configuration state only where relevant to stated warranty conditions. Coverage: warranted assets with verified identities and approved telemetry access. Timing: daily deadline calculation and event capture through the warranty period; aggregate by asset, warranty, condition and notice date. Quality limits: telemetry may be incomplete or outside the warranty measurement definition. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the accepted warranty document and registration record, maintained by the Estates Commercial or Procurement owner with the Asset Information owner. Corroboration: purchase, handover, commissioning, maintenance and supplier contact records. Precedence: the executed warranty terms and authorised commercial interpretation take precedence over summaries or manufacturer marketing. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag unregistered assets, wrong serials, ambiguous start dates, missing maintenance evidence, extensions and conflicting supplier records. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: lookup. Decision cadence: warranty coverage and deadline extraction support, daily and before planned intervention. Minimum evidence: matched asset and warranty, effective start, duration, covered scope, conditions, notice method and authorised source. Method: structured term extraction verified against a human-maintained summary, with deadline rules and exception flags. Uncertainty: date range, interpretation confidence and unresolved clauses. Safe failure: state that commercial confirmation is required when terms, identity or dates are ambiguous.

ANSWER BOUNDARY

Do not provide legal interpretation, promise coverage, submit claims, contact suppliers, disclose commercial terms broadly or delay urgent safe work for warranty reasons. Formal records, competent persons and accountable assurance owners retain authority over compliance and warranty conclusions.

WHY THIS IS USEFUL

Warranty owners can identify potential claims early enough for competent assessment and timely notification.

EA-052

STAKEHOLDER QUESTION

Which recent failures have enough evidence for warranty screening, and what exclusions or missing facts need competent assessment?

SENSORS AND TELEMETRY

Proposed evidence: fault, runtime, operating-limit and post-failure state evidence relevant to the covered failure. Coverage: warranted assets with healthy sensing and a confirmed failure event. Timing: event-level capture with daily candidate screening; aggregate by asset and potential covered event. Quality limits: operating outside a sensor range does not prove misuse, and many warranty exclusions are not measurable. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: warranty terms and validated failure record, maintained by the Estates Commercial owner with Maintenance. Corroboration: work orders, commissioning, maintenance compliance, configuration history, parts, supplier notices and defect evidence. Precedence: authorised commercial and technical owners determine claim suitability; analytics only identifies evidence patterns. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag uncertain failure cause, missed notice periods, incomplete maintenance, modifications, mismatched serials and disputed coverage. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: comparison and recommendation. Decision cadence: warranty-claim candidate triage, daily during active warranty. Minimum evidence: confirmed warranty identity, credible failure, relevant term, timely notice window and evidence of required maintenance. Method: rule matching between event facts and structured terms, with explicit exclusion and ambiguity flags. Uncertainty: candidate strength and missing evidence, not probability of success. Safe failure: route to commercial and technical owners without a positive recommendation when terms or cause are unclear.

ANSWER BOUNDARY

Do not diagnose legal liability, promise reimbursement, submit or reject a claim, delay safety action or conceal evidence that may weaken a claim. Formal records, competent persons and accountable assurance owners retain authority over compliance and warranty conclusions.

WHY THIS IS USEFUL

The warranty owner can assemble candidate evidence without promising claim validity or outcome.

EA-053

STAKEHOLDER QUESTION

Which expected statutory inspection or test records are absent, expired, inaccessible or not linked to the correct asset?

SENSORS AND TELEMETRY

Proposed evidence: due-date reminders, asset identity verification and selected test-result telemetry only when the approved process permits it. Coverage: assets and systems covered by the institution's statutory register. Timing: daily due-status refresh and at completion of an authorised inspection or test; aggregate by obligation, asset and responsible owner. Quality limits: sensor operation cannot substitute for a required competent inspection or formal record. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the current statutory asset and obligation register, maintained by the Estates compliance owner. Corroboration: competent-person reports, certificates, test sheets, asset register, change records and approved exemptions. Precedence: the current compliance owner and competent professional determine applicability and status; formal records take precedence over telemetry. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag wrong asset links, changed applicability, superseded certificates, missing competence evidence, ambiguous extensions and conflicting due dates. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: lookup and comparison. Decision cadence: record completeness and due-status checking, daily with defined escalation windows. Minimum evidence: applicable obligation, asset identity, required record type, competent issuer, effective and expiry dates and current owner. Method: rule-based due calculation plus identity, scope and document-status validation. Uncertainty: present, due soon, overdue, unverified or not applicable, with reasons. Safe failure: return unverified and escalate rather than infer compliance where applicability or competent evidence is uncertain.

ANSWER BOUNDARY

Do not certify compliance, determine legal applicability independently, extend due dates, approve exemptions or replace competent inspections. Formal records, competent persons and accountable assurance owners retain authority over compliance and warranty conclusions.

WHY THIS IS USEFUL

Assurance owners can locate missing or expired formal evidence and arrange competent action.

EA-054

STAKEHOLDER QUESTION

Are certificates and formal test records attached to the correct physical asset, current configuration and applicable record version?

SENSORS AND TELEMETRY

Proposed evidence: tag and serial verification, configuration identifiers and timestamped test-result capture where approved. Coverage: assets with formal inspection or test records. Timing: at record receipt and after asset or configuration change; aggregate as record-to-asset link confidence. Quality limits: shared systems, replaced components and generic certificates can make linkage ambiguous. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the formal inspection or test record, maintained by the Estates compliance owner and competent record issuer. Corroboration: asset register, configuration history, project changes, commissioning, location and predecessor-successor links. Precedence: the competent issuer's scoped record and current asset identity take precedence; filename or folder location is not sufficient. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag generic asset descriptions, duplicate serials, changed configuration, wrong locations, superseded versions and missing scope. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: comparison. Decision cadence: record-to-asset and version validation, on ingestion and monthly for changed assets. Minimum evidence: record issuer, date, scope, asset identifier or serial, configuration context and current record status. Method: deterministic identity checks, predecessor-successor reasoning and version supersession rules. Uncertainty: matched, probable, ambiguous or mismatched with traceable reasons. Safe failure: keep the record unattached or provisionally linked and request issuer or owner confirmation when identity is not secure.

ANSWER BOUNDARY

Do not alter the record, certify its validity, infer coverage beyond its scope or disclose restricted certificates to unauthorised users. Formal records, competent persons and accountable assurance owners retain authority over compliance and warranty conclusions.

WHY THIS IS USEFUL

The assurance owner can correct document-to-asset links before a valid record is relied on for the wrong configuration.

EA-055

STAKEHOLDER QUESTION

Which asset records are restricted, and what level of detail is permitted for each current role and purpose?

SENSORS AND TELEMETRY

Proposed evidence: classification metadata for feeds such as precise location, security state, low-count occupancy or control topology. Coverage: all asset data sources and generated outputs. Timing: at onboarding and whenever purpose, source or role changes; aggregate by field and role, with minimum necessary disclosure. Quality limits: automatic classification can miss contextual sensitivity and does not grant permission. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the institutional information-classification and access policy, maintained by the Estates Information Governance and Security owners. Corroboration: source-owner permissions, contract restrictions, privacy assessments, research restrictions and system roles. Precedence: explicit current owner permission and policy take precedence over inherited system access or previous exports. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag stale role membership, copied files, mixed-classification documents, unclear purpose and conflicting source restrictions. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: lookup. Decision cadence: field-level access and redaction evaluation, at every query and permission change. Minimum evidence: authenticated role, approved purpose, field classification, source permission and requested output granularity. Method: deny-by-default policy rules, location masking, low-count suppression and redaction logging. Uncertainty: allowed, aggregated, redacted or denied fields with reason code. Safe failure: deny the restricted part and offer a safer aggregate or owner contact when permission is incomplete.

ANSWER BOUNDARY

Do not broaden access, infer entitlement from job title alone, expose precise sensitive locations, redistribute supplier-restricted content or reveal individual-level data. Formal records, competent persons and accountable assurance owners retain authority over compliance and warranty conclusions.

WHY THIS IS USEFUL

Information owners can enforce minimum-necessary disclosure while preserving access needed for estate work.

EA-056

STAKEHOLDER QUESTION

Where do certificates or formal records conflict with the asset's current status, serial, location or configuration?

SENSORS AND TELEMETRY

Proposed evidence: current tag, serial, configuration and service-state verification where safe and permitted. Coverage: assets with a formal record and a current operational identity. Timing: on record ingestion, asset change and monthly exception checks; aggregate by conflict field and consequence. Quality limits: field observations can be wrong or reflect a later legitimate change not yet documented. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the formal record and current asset master, maintained by the Estates compliance and Asset Information owners. Corroboration: project changes, commissioning, replacement history, location records and issuer correspondence. Precedence: competent record scope and current approved asset identity remain separate authorities; conflicts require owner resolution. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag post-record modifications, swapped components, transcription errors, duplicate serials, changed room identifiers and superseded records. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: comparison. Decision cadence: formal-record consistency checking, monthly and before reliance on the record. Minimum evidence: record version and scope, current asset identity and configuration, change history and observation dates. Method: field-by-field comparison with temporal reasoning to distinguish later change from original mismatch. Uncertainty: conflict severity, likely temporal explanation and confidence. Safe failure: mark the record as requiring confirmation and exclude it from assurance status where the conflict affects scope.

ANSWER BOUNDARY

Do not invalidate a certificate, certify the asset, change identity or interpret legal consequence. Formal records, competent persons and accountable assurance owners retain authority over compliance and warranty conclusions.

WHY THIS IS USEFUL

The competent issuer and assurance owner can resolve mismatches before the record supports a decision.

EA-057

STAKEHOLDER QUESTION

Which assurance, inspection, warranty or competence requirements may change when an asset is modified, replaced, relocated or used differently?

SENSORS AND TELEMETRY

Proposed evidence: configuration changes, duty changes, new load patterns and service relationships that are observable in approved systems. Coverage: changed assets and spaces in the authorised change scope. Timing: at change proposal, commissioning and accepted use change; aggregate by potential obligation trigger. Quality limits: telemetry cannot determine legal applicability or all hidden modifications. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the approved change description and obligation applicability rules, maintained by the Estates compliance owner with competent professionals. Corroboration: asset specifications, space-use approvals, statutory register, risk assessments, design records and formal guidance. Precedence: competent and authorised applicability decisions take precedence; analytics only identifies possible triggers. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag incomplete change scope, temporary use, local modifications, outdated rule mappings. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: comparison and forecast. Decision cadence: change-to-obligation trigger screening, at each change gate. Minimum evidence: defined before-and-after state, current obligation set, affected asset or use attributes and competent owner for each domain. Method: transparent rule mapping from changed attributes to possible new, changed or retired obligations. Uncertainty: possible trigger with rule source and uncertainty. Safe failure: state that applicability is unconfirmed and require competent assessment whenever a material trigger cannot be ruled out.

ANSWER BOUNDARY

Do not determine legal obligations, approve a change, waive an inspection, certify compliance or infer current room use from the historical plan. Formal records, competent persons and accountable assurance owners retain authority over compliance and warranty conclusions.

WHY THIS IS USEFUL

Project teams can bring potential assurance consequences into planning before the modified asset returns to service.

EA-058

STAKEHOLDER QUESTION

What operational telemetry can corroborate assurance evidence without being mistaken for a formal inspection, test or certificate?

SENSORS AND TELEMETRY

Proposed evidence: continuous equipment state, alarms, environmental conditions, test-point readings and sensor-health evidence selected for the obligation. Coverage: only the assets and parameters explicitly mapped to the assurance question. Timing: at engineering-appropriate sampling with daily or monthly summaries and retained exceptions; aggregate by asset and assurance criterion. Quality limits: continuous monitoring may be indirect, calibrated differently or unable to observe required physical conditions. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the formal assurance requirement and accepted evidence definition, maintained by the Estates compliance owner and competent professional. Corroboration: inspection or test records, calibration, commissioning, maintenance history, control logic and approved operating limits. Precedence: formal competent evidence governs assurance status; telemetry is supplementary and cannot extend a certificate or inspection scope. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag uncalibrated sensors, incomplete coverage, gaps, changed configuration, ambiguous thresholds and conflicts with formal records. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: observation and comparison. Decision cadence: supplementary-evidence mapping, continuous monitoring with reporting at the authorised interval. Minimum evidence: defined criterion, mapped signal, calibration or health, coverage, threshold basis and current formal record. Method: evidence-chain evaluation distinguishing direct, indirect and contextual support. Uncertainty: support strength, coverage and periods not observed. Safe failure: show no assurance conclusion when telemetry is not an accepted measure or its quality is insufficient.

ANSWER BOUNDARY

Do not label telemetry as certification, declare compliance, replace inspection, infer safety or suppress contrary formal evidence. Formal records, competent persons and accountable assurance owners retain authority over compliance and warranty conclusions.

WHY THIS IS USEFUL

Competent professionals can use telemetry to target attention between formal checks without treating it as certification.

EA-059

STAKEHOLDER QUESTION

Which records or conclusions require competent-person confirmation before an estate decision can rely on them?

SENSORS AND TELEMETRY

Proposed evidence: evidence-quality flags such as missing calibration, contradictory values, inaccessible asset identity or changed configuration. Coverage: records feeding safety, compliance, structural, electrical, mechanical or other competence-dependent decisions. Timing: at retrieval for the decision and after material change; aggregate by record, decision and required competence domain. Quality limits: automated quality checks cannot assess professional competence or hidden physical conditions. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the institutional competence and evidence-acceptance rules, maintained by the Estates compliance and professional governance owners. Corroboration: record issuer details, scope, accreditation or appointment evidence, inspection date, asset identity and change history. Precedence: current governance and the relevant competent professional determine acceptability; document presence alone has no precedence. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag missing issuer role, expired appointment, out-of-scope statements, post-record changes and conflicting evidence. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: recommendation. Decision cadence: competence-gate and reliance screening, at every high-consequence decision. Minimum evidence: decision type, required competence, record scope, issuer role, date, asset match and current configuration. Method: rule-based gate checks with explicit escalation to the relevant professional domain. Uncertainty: eligible, conditional or confirmation required, with reasons. Safe failure: withhold the record from the decision evidence set where competence or scope cannot be verified.

ANSWER BOUNDARY

Do not judge professional competence beyond approved records, certify the evidence, interpret liability or make the dependent decision. Formal records, competent persons and accountable assurance owners retain authority over compliance and warranty conclusions.

WHY THIS IS USEFUL

Decision-makers can identify where competent confirmation is a prerequisite rather than treating document presence as validity.

EA-060

STAKEHOLDER QUESTION

What warranty notices, inspections, tests, records or owner confirmations are due soon, and which dependencies could prevent timely completion?

SENSORS AND TELEMETRY

Proposed evidence: event and usage data only where it legitimately changes a due trigger, such as run hours under an approved maintenance or warranty rule. Coverage: assets with current obligations and matched identities. Timing: daily refresh with owner-defined reminder windows; aggregate by action, due window, asset and responsible role. Quality limits: missing runtime, counter resets and ambiguous trigger rules can make usage-based dates uncertain. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the authoritative obligation, warranty and action schedules, maintained by the Estates compliance, commercial and maintenance owners. Corroboration: formal terms, inspection records, maintenance plans, project changes, authorised extensions and owner assignments. Precedence: current authorised due dates and formal terms take precedence; calculated dates are advisory until owner-confirmed. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag missing start dates, superseded obligations, unrecorded extensions, unmatched assets and conflicts between calendar and usage triggers. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: lookup and forecast. Decision cadence: deadline and trigger calculation, daily and on relevant state change. Minimum evidence: current obligation, trigger definition, effective start or last completion, asset identity, owner and any authorised exception. Method: calendar and usage-based rules with business-day handling and confidence checks. Uncertainty: due date range, reminder status and assumptions. Safe failure: show a due-date uncertainty and escalate before the earliest plausible date when required inputs conflict.

ANSWER BOUNDARY

Do not extend deadlines, send statutory or warranty notices, certify completion, authorise work or interpret legal effect. Formal records, competent persons and accountable assurance owners retain authority over compliance and warranty conclusions.

WHY THIS IS USEFUL

Accountable owners can arrange competent action, evidence and notification before a current deadline is missed.

EA-061

STAKEHOLDER QUESTION

What is the evidence-based whole-life cost range for continuing the current maintenance approach compared with a defined replacement option?

SENSORS AND TELEMETRY

Proposed evidence: runtime, energy, demand, failure, condition and service-interruption evidence relevant to cost drivers. Coverage: the asset and affected system over a declared analysis boundary. Timing: monthly cost-driver refresh using representative annual and peak periods; aggregate by option and lifecycle phase. Quality limits: future duty, tariffs, failure rates and replacement performance are uncertain. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the current finance ledger, asset costs and approved option assumptions, maintained by the Estates Finance and accountable asset owners. Corroboration: work orders, contracts, spares, energy tariffs, condition surveys, project estimates, residual life and disposal requirements. Precedence: authorised finance records and approved cost assumptions take precedence; telemetry only supplies measured drivers. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag misallocated invoices, bundled contracts, inflation assumptions, incomplete downtime cost and optimism in replacement estimates. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: calculation, estimate and comparison. Decision cadence: discounted lifecycle cost comparison, at option appraisal and when major assumptions change. Minimum evidence: defined options, analysis period, current cost baseline, credible intervention costs, failure scenarios, discount and inflation rules. Method: cash-flow modelling with low, central and high cases plus sensitivity and break-even analysis. Uncertainty: cost ranges, dominant assumptions and probability-free scenario spread unless probabilities are approved. Safe failure: return an evidence-gap model without recommendation when costs or remaining-life assumptions are not credible.

ANSWER BOUNDARY

Do not approve investment, procurement or disposal, create an accounting valuation, guarantee savings or replace finance and quantity-surveying evidence. Finance, procurement, valuation and accountable asset owners retain authority over expenditure, value and disposal decisions.

WHY THIS IS USEFUL

Decision-makers can see the cost range, assumptions and evidence needed for a governed maintain-or-replace appraisal.

EA-062

STAKEHOLDER QUESTION

Which assets or systems have unusually high energy or maintenance spend relative to the service and duty they provide?

SENSORS AND TELEMETRY

Proposed evidence: submetered energy, runtime, load, delivered output proxies and service availability. Coverage: assets with valid metering and a defined service denominator; shared plant remains explicitly allocated or unallocated. Timing: 15-minute or hourly capture aggregated monthly and seasonally; aggregate by asset class, service unit and duty band. Quality limits: meter boundaries, weather, occupancy, research load and shared services can distort comparison. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: energy invoices and maintenance cost records, maintained by the Estates Energy and Finance owners. Corroboration: meter maps, work orders, asset hierarchy, service output definitions, tariffs, space use and weather data. Precedence: finance and meter owners govern cost and consumption; service owners govern the output denominator. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag estimated bills, missing submetering, cost booked to parent assets, changed service levels and inconsistent peer definitions. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: calculation and comparison. Decision cadence: cost-and-energy intensity benchmarking, monthly and annually. Minimum evidence: verified meter boundary, attributable cost, service denominator, comparable duty and at least one representative period. Method: normalised intensity and peer distribution with weather or load adjustment where justified. Uncertainty: attribution range, peer comparability and unallocated share. Safe failure: avoid ranking when denominators or meter boundaries are not comparable; show the data gap instead.

ANSWER BOUNDARY

Do not label a service wasteful, infer user behaviour, promise savings, approve replacement or allocate charges beyond approved finance rules. Finance, procurement, valuation and accountable asset owners retain authority over expenditure, value and disposal decisions.

WHY THIS IS USEFUL

Energy and asset teams can select systems for engineering investigation, metering enhancement or option appraisal.

EA-063

STAKEHOLDER QUESTION

Which maintenance, energy and project costs can be attributed confidently to this asset or system, and which should remain shared or unattributed?

SENSORS AND TELEMETRY

Proposed evidence: meter-boundary readings, runtime and work-event timestamps that support but do not determine allocation. Coverage: the selected asset, parent system and shared-service boundaries. Timing: monthly and at project close; aggregate as direct, rule-allocated, estimated or unallocated cost. Quality limits: telemetry cannot identify every invoice line, shared overhead or contractual allocation. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the finance ledger and approved coding structure, maintained by the Estates Finance owner. Corroboration: work orders, purchase orders, contracts, meter maps, asset hierarchy, project codes and allocation policy. Precedence: authorised finance coding and policy take precedence; operational data may challenge a link but cannot post or reallocate cost. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag generic cost centres, bundled invoices, wrong asset links, parent-level booking, late accruals and conflicting project codes. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: calculation and estimate. Decision cadence: cost lineage and attribution classification, monthly and at decision gate. Minimum evidence: transaction identifier, amount, period, approved code, asset or system relationship and allocation rule where indirect. Method: lineage tracing from transaction to work, project or meter, with residual unallocated balances. Uncertainty: attribution class, confidence and total unallocated proportion. Safe failure: leave costs unallocated and state the missing link when a defensible rule does not exist.

ANSWER BOUNDARY

Do not amend ledgers, create charges, infer fraud, produce valuations or force exact asset attribution for shared costs. Finance, procurement, valuation and accountable asset owners retain authority over expenditure, value and disposal decisions.

WHY THIS IS USEFUL

Finance and asset teams can separate decision-ready attribution from shared or uncertain cost.

EA-064

STAKEHOLDER QUESTION

What is the credible range for this asset's remaining useful life, and which condition, exposure or obsolescence assumptions drive it?

SENSORS AND TELEMETRY

Proposed evidence: condition trends, run hours, starts, load, environment and failure events relevant to known degradation modes. Coverage: the asset and comparable cohort, while recognising unobservable condition mechanisms. Timing: monthly trend update and at major condition or duty change; aggregate as a range, not an exact retirement date. Quality limits: limited history, sensor changes and hidden deterioration can dominate uncertainty. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: current condition assessment and asset history, maintained by the Estates condition and asset strategy owners. Corroboration: installation and refurbishment dates, manufacturer guidance, peer failures, work history, duty, environment and planned changes. Precedence: competent condition judgement and approved asset strategy assumptions take precedence over statistical estimates. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag unknown installation date, replaced components, survivor bias, changed duty, weak peer data and obsolete manufacturer assumptions. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: estimate and forecast. Decision cadence: remaining-life range estimation, annually and at investment appraisal. Minimum evidence: credible age or component history, current condition, relevant duty and failure data, and an explicit definition of end of useful life. Method: engineering judgement structured with survival or degradation models where sample support exists, plus scenario sensitivity. Uncertainty: percentile or low-central-high range and confidence drivers. Safe failure: state that remaining life is not estimable and specify the survey or history needed when evidence is inadequate.

ANSWER BOUNDARY

Do not certify condition, create an accounting life, promise operation to a date, approve disposal or replace competent engineering judgement. Finance, procurement, valuation and accountable asset owners retain authority over expenditure, value and disposal decisions.

WHY THIS IS USEFUL

Lifecycle planners can use a range and target the evidence most likely to change replacement timing.

EA-065

STAKEHOLDER QUESTION

Which defined intervention options reduce the greatest critical risk or backlog exposure within a stated budget and delivery constraint?

SENSORS AND TELEMETRY

Proposed evidence: current condition, availability, energy and demand evidence that informs option effects. Coverage: candidate assets and services, with restricted risk detail limited to authorised decision roles. Timing: monthly evidence refresh and at each portfolio option set; aggregate by option, risk domain and budget year. Quality limits: option benefits may depend on untested assumptions, future demand and interactions with other projects. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: approved backlog, risk, cost and investment criteria, maintained by the Estates Strategy, Finance and institutional risk owners. Corroboration: condition surveys, project estimates, service consequences, dependencies, carbon effects, deliverability and funding constraints. Precedence: accountable owners set risk appetite, option eligibility and budget; analytics only compares declared assumptions. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag double-counted benefits, incomparable risk scales, optimistic cost or delivery, missing dependencies and disputed weights. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: comparison and recommendation. Decision cadence: constrained portfolio option analysis, at the capital-planning cycle and when constraints change. Minimum evidence: eligible options, cost ranges, risk-reduction evidence, dependencies, delivery capacity and approved decision criteria. Method: multi-criteria and knapsack-style scenario analysis with sensitivity, dominance and portfolio interaction checks. Uncertainty: sets of robust candidates rather than one deterministic winner. Safe failure: present alternative portfolios and unresolved assumptions; do not recommend a single approval when rankings are unstable.

ANSWER BOUNDARY

Do not approve investment, set the budget, create valuations, promise risk reduction or substitute for finance, procurement and accountable-owner decisions. Finance, procurement, valuation and accountable asset owners retain authority over expenditure, value and disposal decisions.

WHY THIS IS USEFUL

Leaders can compare robust candidate portfolios while retaining accountable investment and budget decisions.

EA-066

STAKEHOLDER QUESTION

What cost and service exposure could arise from downtime, disruption or deferring intervention under defined scenarios?

SENSORS AND TELEMETRY

Proposed evidence: service unavailability, aggregate affected demand, environmental excursion and restoration-time evidence. Coverage: the affected services and declared disruption boundary; no individual activity or research-content monitoring. Timing: event capture with monthly and scenario-based aggregation; aggregate by consequence category and time window. Quality limits: unrecorded workarounds, reputational effects and research losses may be difficult or inappropriate to monetise. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: approved service consequence and finance assumptions, maintained by the Estates Finance and service owners. Corroboration: incident records, continuity plans, timetable or booking impacts, temporary accommodation, contractor costs and backlog deferral assumptions. Precedence: service owners determine consequences and Finance determines accepted monetisation; telemetry cannot create financial loss values. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag missing disruption logs, double-counted impacts, unapproved shadow prices, uncertain restoration and sensitive research consequences. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: estimate and forecast. Decision cadence: bounded disruption and deferral cost scenario, at option appraisal and after significant events. Minimum evidence: defined failure or deferral scenario, duration range, affected services, approved direct-cost rates and explicitly accepted non-financial consequences. Method: time-based scenario model separating direct, contingent and non-monetised effects. Uncertainty: low-central-high direct cost and a separate qualitative consequence statement. Safe failure: leave effects unmonetised and state uncertainty where no approved value or defensible evidence exists.

ANSWER BOUNDARY

Do not invent loss values, expose confidential research impact, promise avoided cost, approve intervention or create accounting provisions. Finance, procurement, valuation and accountable asset owners retain authority over expenditure, value and disposal decisions.

WHY THIS IS USEFUL

Option appraisals can include service and deferral exposure without converting uncertain impacts into booked loss.

EA-067

STAKEHOLDER QUESTION

Which assets have low or zero recorded book value but high operational consequence, replacement complexity or dependency risk?

SENSORS AND TELEMETRY

Proposed evidence: service availability, dependency and aggregate demand evidence. Coverage: assets with current finance and criticality links. Timing: monthly and before disposal or replacement planning; aggregate by consequence domain and finance value band. Quality limits: book value is an accounting measure, while operational use and redundancy can change quickly. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the finance asset record and approved criticality assessment, maintained by the Estates Finance and accountable asset owners. Corroboration: service dependencies, business continuity, condition, replacement lead time, spares and work history. Precedence: Finance governs recorded value; service and risk owners govern consequence; neither should overwrite the other. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag fully depreciated assets, missing finance links, parent-level values, unverified redundancy and stale criticality. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: comparison. Decision cadence: value-consequence divergence detection, monthly and at portfolio planning. Minimum evidence: matched finance asset, current recorded value, confirmed service dependency, criticality and contingency evidence. Method: two-axis classification with confidence and sensitivity to disputed criticality or redundancy. Uncertainty: divergence band and evidence completeness. Safe failure: return an unmatched or provisional item when finance and operational identities cannot be reconciled.

ANSWER BOUNDARY

Do not create a valuation, infer that low book value means disposal, approve replacement or disclose sensitive service dependencies broadly. Finance, procurement, valuation and accountable asset owners retain authority over expenditure, value and disposal decisions.

WHY THIS IS USEFUL

Planning can protect operationally essential assets that financial depreciation alone would understate.

EA-068

STAKEHOLDER QUESTION

What evidence supports reuse, transfer, resale, component recovery or disposal routes when an asset leaves service?

SENSORS AND TELEMETRY

Proposed evidence: final operating condition, runtime, configuration and hazardous-state observations only where safe and relevant. Coverage: outgoing assets approved for option assessment. Timing: at pre-decommission survey and after safe removal; aggregate by asset and possible route. Quality limits: remote data cannot establish physical condition, title, contamination, market demand or safe reuse. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: asset ownership, disposal and environmental requirements, maintained by the Estates Asset, Finance, Procurement and Sustainability owners. Corroboration: condition survey, maintenance history, warranties, licences, hazardous-material information, market evidence, project needs and authorised waste records. Precedence: ownership, finance, safety and procurement rules take precedence; competent assessment is required for reuse or disposal suitability. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag unknown title, restricted software, missing hazardous information, obsolete specifications, uncertain condition and changing market evidence. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: comparison and recommendation. Decision cadence: end-of-life route evidence screening, at option development and before disposal action. Minimum evidence: confirmed ownership, current condition assessment, restrictions, potential recipient or market route, removal cost and required environmental evidence. Method: eligibility rules with low-central-high net-cost scenarios and non-financial constraints. Uncertainty: feasible, conditional or unassessable routes with missing evidence. Safe failure: state that a route is only a candidate and require authorised professional and commercial confirmation.

ANSWER BOUNDARY

Do not approve disposal, advertise or sell an asset, certify reuse safety, create residual value or handle controlled waste decisions. Finance, procurement, valuation and accountable asset owners retain authority over expenditure, value and disposal decisions.

WHY THIS IS USEFUL

Owners can commission the checks needed for a viable circular route before disposal decisions are made.

EA-069

STAKEHOLDER QUESTION

Which assumptions in an asset investment case are well supported, weak, stale or internally inconsistent?

SENSORS AND TELEMETRY

Proposed evidence: measured demand, energy, condition, failure, runtime and service-state evidence linked to cited assumptions. Coverage: the assets and services inside the business-case boundary. Timing: at each option iteration and before decision submission; aggregate by assumption and decision sensitivity. Quality limits: historical evidence may not represent future operating conditions or proposed technology performance. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the current business-case model and approved source pack, maintained by the Estates Strategy and Finance owners. Corroboration: project estimates, surveys, asset records, tariffs, carbon factors, maintenance costs, demand forecasts, risk and service-owner evidence. Precedence: authorised finance and governance assumptions take precedence; all derived values must retain source lineage. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag copied assumptions, mixed price bases, outdated tariffs, double-counted benefits, hidden contingencies and conflicting analysis periods. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: comparison. Decision cadence: assumption assurance and sensitivity analysis, at every material model update. Minimum evidence: assumption value, unit, price or time base, source, owner, uncertainty range and dependent outputs. Method: lineage, unit and consistency checks plus one-at-a-time and scenario sensitivity. Uncertainty: supported, conditional, weak or conflicting and the decision impact of each. Safe failure: withhold a definitive option ranking when influential assumptions lack defensible ranges or ownership.

ANSWER BOUNDARY

Do not approve the case, create valuations, select optimistic assumptions, guarantee benefits or replace finance and professional sign-off. Finance, procurement, valuation and accountable asset owners retain authority over expenditure, value and disposal decisions.

WHY THIS IS USEFUL

The case owner can target evidence collection at assumptions capable of changing the preferred option.

EA-070

STAKEHOLDER QUESTION

How could related intervention candidates be grouped and sequenced to avoid duplicated enabling work, incompatible shutdowns or stranded assets?

SENSORS AND TELEMETRY

Proposed evidence: current load, service state, condition and capacity evidence that constrains sequence. Coverage: candidate assets, enabling infrastructure and affected services across the declared portfolio scope. Timing: monthly refresh and at programme option updates; aggregate by dependency, geographic work package and delivery window. Quality limits: future projects, funding and technical interfaces may change, and telemetry cannot define construction feasibility. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the approved project pipeline, asset strategy and dependency records, maintained by the Estates Strategy and Capital Projects owners. Corroboration: condition, backlog, decarbonisation plans, space changes, outages, design lives, cost estimates and procurement constraints. Precedence: accountable programme owners determine eligibility and sequence; analytics highlights dependency and timing consequences. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag provisional projects, inconsistent boundaries, omitted enabling works, uncertain funding, incompatible design assumptions and conflicting dates. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: forecast and recommendation. Decision cadence: dependency-aware programme sequencing, at annual planning and when a major project changes. Minimum evidence: candidate scope, costs, dependencies, useful-life constraints, service windows, delivery capacity and approved objectives. Method: precedence graph, clash detection and scenario scheduling with stranded-work indicators. Uncertainty: multiple feasible sequences, dependency confidence and sensitivity to dates. Safe failure: return infeasible or unresolved sequence sets when dependencies or constraints are incomplete.

ANSWER BOUNDARY

Do not approve a programme, commit funds, procure works, certify feasibility or promise delivery dates. Finance, procurement, valuation and accountable asset owners retain authority over expenditure, value and disposal decisions.

WHY THIS IS USEFUL

Programme planners can identify sensible bundles and sequences for governed feasibility and funding decisions.

EA-071

STAKEHOLDER QUESTION

Which assets and services are most exposed to power, heat, flood, water-supply or communications disruption, and how current is the evidence?

SENSORS AND TELEMETRY

Proposed evidence: power quality, environmental conditions, leak or flood detection, water pressure or flow, network-control health and equipment-state telemetry. Coverage: declared assets and services, with sensitive resilience details restricted by role. Timing: near real time for current exposure and seasonal or annual scenario summaries; aggregate by hazard, service and dependency. Quality limits: sensor placement is incomplete, hazard intensity may exceed observed ranges and shared external dependencies may be hidden. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the approved hazard, asset and dependency records, maintained by the Estates risk and Business Continuity owners. Corroboration: site assessments, utility information, flood or climate data, system topology, criticality, condition, contingency tests and incident history. Precedence: current institutional risk and competent engineering assessments take precedence over automated vulnerability scores. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag outdated hazard layers, unverified routes, changed site drainage, missing external dependencies and untested mitigations. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: comparison and estimate. Decision cadence: multi-hazard vulnerability screening, seasonally, annually and after major change. Minimum evidence: hazard exposure, asset susceptibility, service consequence, redundancy or protection and current evidence date. Method: hazard-asset-service graph analysis with scenario bands and mitigation checks. Uncertainty: vulnerability band, confidence and unmodelled hazards. Safe failure: return a screening gap and require competent site or system assessment when exposure or protection is unverified.

ANSWER BOUNDARY

Do not certify resilience, disclose sensitive dependency maps broadly, predict exact damage, authorise emergency action or replace specialist hazard assessment. Accountable portfolio owners retain authority over resilience, carbon and investment pathways; scenarios are not forecasts or commitments.

WHY THIS IS USEFUL

Resilience teams can target specialist assessment and adaptation at evidence-backed exposure and consequence.

EA-072

STAKEHOLDER QUESTION

Under a defined outage scenario, which critical services could continue, at what demand and for what credible duration?

SENSORS AND TELEMETRY

Proposed evidence: current load, UPS or generator state, battery or fuel indicators, standby equipment availability and service-state telemetry. Coverage: declared critical services and authorised resilience assets. Timing: near real time for current state, with scenario calculation on request; aggregate by service and outage duration band. Quality limits: nameplate capacity, fuel level and standby status do not prove successful changeover or sustained performance. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the approved continuity and emergency-power arrangements, maintained by the Estates engineering and Business Continuity owners. Corroboration: load studies, autonomy tests, fuel contracts, maintenance status, critical service mappings, operating procedures and current outages. Precedence: competent test evidence and authorised continuity plans take precedence; live telemetry only constrains current readiness. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag untested batteries, uncertain load growth, shared circuits, unavailable staff, stale fuel assumptions and incomplete dependency maps. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: estimate and forecast. Decision cadence: bounded continuity-duration scenario, for current state and selected planning scenarios. Minimum evidence: verified supported load, usable capacity, current availability, tested efficiency, fuel or battery state and dependency sequence. Method: energy or resource balance with low-central-high demand and derating scenarios. Uncertainty: duration range and the assumptions that terminate continuity. Safe failure: state that duration is unverified and escalate when topology, load or test evidence is insufficient.

ANSWER BOUNDARY

Do not promise continuity, initiate emergency systems, authorise load shedding, disclose sensitive resilience information widely or certify emergency readiness. Accountable portfolio owners retain authority over resilience, carbon and investment pathways; scenarios are not forecasts or commitments.

WHY THIS IS USEFUL

Contingency owners can identify credible autonomy, limiting dependencies and tests still required.

EA-073

STAKEHOLDER QUESTION

Which proposed decarbonisation measures interact with existing condition, capacity, control or service constraints?

SENSORS AND TELEMETRY

Proposed evidence: energy, peak demand, temperatures, flows, control positions, runtime and environmental performance relevant to the measure. Coverage: candidate systems and affected zones over representative seasons. Timing: 15-minute or hourly capture, aggregated monthly and seasonally; aggregate by option, constraint and operating condition. Quality limits: historical operation may be inefficient, sensors may not cover design extremes and future controls may behave differently. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the current decarbonisation option definition and asset condition or capacity records, maintained by the Estates Sustainability, Energy and engineering owners. Corroboration: surveys, system schematics, design criteria, BMS strategy, electrical capacity, space constraints, projects and maintenance backlog. Precedence: competent design and current asset evidence take precedence; analytics is an early constraint screen. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag unverified plant capacity, hidden distribution limitations, outdated controls documentation, temporary faults and incomplete option boundaries. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: comparison and estimate. Decision cadence: option-to-existing-system constraint screening, at concept and option development. Minimum evidence: defined measure, current topology, condition, capacity, controls, representative demand and known planned changes. Method: dependency and capacity checks under design-day and part-load scenarios with explicit survey gaps. Uncertainty: constraint severity and confidence by option. Safe failure: return a requirement for survey, modelling or trial when current evidence cannot support compatibility.

ANSWER BOUNDARY

Do not approve the measure, certify design performance, promise carbon savings, control plant or replace professional design and commissioning. Accountable portfolio owners retain authority over resilience, carbon and investment pathways; scenarios are not forecasts or commitments.

WHY THIS IS USEFUL

Decarbonisation teams can identify enabling work and avoid options that ignore weak assets or control constraints.

EA-074

STAKEHOLDER QUESTION

Where are the portfolio's highest operational energy and carbon demands, and how confident is the reporting boundary and baseline?

SENSORS AND TELEMETRY

Proposed evidence: fiscal and submetered electricity, heat, fuel and renewable generation plus relevant runtime and weather observations. Coverage: buildings, systems and end uses with verified meter boundaries; unmetered consumption remains explicit. Timing: half-hourly or hourly capture with monthly and annual aggregation; aggregate by building, end use and reporting period. Quality limits: estimated bills, meter gaps, tenant loads, export, degree-day effects and boundary changes affect comparability. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the authorised energy and carbon reporting dataset, maintained by the Estates Energy and Sustainability owners. Corroboration: utility invoices, meter maps, emissions factors, building changes, space or use context and data-quality records. Precedence: approved reporting boundaries, factors and finance-grade consumption take precedence over informal dashboards. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag estimated reads, changed factors, missing meters, duplicated supply points, incomplete renewables data and conflicting periods. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: calculation and comparison. Decision cadence: energy and operational-carbon baseline construction, monthly and annually. Minimum evidence: verified boundary, consumption or generation, period, factor version, weather and material change context. Method: meter hierarchy reconciliation, gap estimation kept separate from measured data, weather normalisation and uncertainty propagation. Uncertainty: measured and estimated shares plus carbon range. Safe failure: withhold building or end-use ranking where boundaries or estimated shares make comparison unreliable.

ANSWER BOUNDARY

Do not certify a carbon account, promise reductions, infer occupant behaviour, allocate tenant charges or hide estimated data inside measured totals. Accountable portfolio owners retain authority over resilience, carbon and investment pathways; scenarios are not forecasts or commitments.

WHY THIS IS USEFUL

Portfolio teams can target metering, validation and option development while keeping measured and estimated totals distinct.

EA-075

STAKEHOLDER QUESTION

Which asset classes may become obsolete, unsupported or unsuitable under plausible technical, regulatory, cybersecurity or service changes?

SENSORS AND TELEMETRY

Proposed evidence: current utilisation, performance margin, firmware or support status and environmental duty where relevant. Coverage: assets in the selected planning horizon. Timing: quarterly evidence refresh with annual scenario assessment; aggregate by obsolescence driver and time horizon. Quality limits: vendor roadmaps and future rules can change, while low current use does not prove future redundancy. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the approved asset strategy and scenario assumptions, maintained by the Estates Strategy owner. Corroboration: vendor support notices, parts availability, standards guidance, planned service changes, decarbonisation pathways, projects and condition history. Precedence: current official notices, competent interpretation and accountable scenario owners take precedence over speculative signals. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag unconfirmed end-of-support dates, supplier dependence, changing standards, uncertain service demand and missing alternatives. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: comparison and forecast. Decision cadence: obsolescence exposure screening, annually and on confirmed vendor or policy change. Minimum evidence: asset identity, dependency, support or requirement change, time horizon, alternatives and consequence of delayed action. Method: driver-based scenario mapping with lead-time and dependency checks. Uncertainty: possible exposure window and confidence in the trigger. Safe failure: classify as watch, investigate or unconfirmed rather than assign an exact obsolescence date.

ANSWER BOUNDARY

Do not provide legal interpretation, declare an asset non-compliant, approve replacement, forecast vendor behaviour as fact or create procurement commitments. Accountable portfolio owners retain authority over resilience, carbon and investment pathways; scenarios are not forecasts or commitments.

WHY THIS IS USEFUL

Owners can create an early-warning list for spares, skills, migration and support planning.

EA-076

STAKEHOLDER QUESTION

How would plausible growth or change in teaching, research and workplace demand affect asset capacity, service headroom and operational risk?

SENSORS AND TELEMETRY

Proposed evidence: aggregate occupancy, booked demand, energy, ventilation, thermal, electrical and water loads. Coverage: spaces and systems included in the scenario; no inspection of research content, individual schedules or productivity. Timing: representative term-time and seasonal intervals with scenario-specific peaks; aggregate by service, zone and demand band. Quality limits: future programmes, equipment, behaviour and simultaneous peaks are uncertain. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the authorised demand scenario and current capacity evidence, maintained by the institutional planning, Space Management and Estates engineering owners. Corroboration: timetables, space allocations, project pipeline, service criteria, load studies, asset condition and continuity constraints. Precedence: approved planning assumptions and competent capacity assessments take precedence over extrapolated historical patterns. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag unapproved growth assumptions, changed room use, hidden specialist loads, incomplete peak coincidence and outdated capacities. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: forecast. Decision cadence: demand-growth capacity scenario, at strategy and project option points. Minimum evidence: defined low, central and high demand, current topology, verified capacity, condition derating and planned changes. Method: peak and annual demand modelling with diversity ranges and service dependency checks. Uncertainty: capacity margin range and tipping conditions. Safe failure: return no capacity conclusion and specify required survey or design study where current evidence is weak.

ANSWER BOUNDARY

Do not infer research plans, identify users, certify capacity, approve growth or promise service performance. Accountable portfolio owners retain authority over resilience, carbon and investment pathways; scenarios are not forecasts or commitments.

WHY THIS IS USEFUL

Capacity owners can identify where demand scenarios justify metering, modelling or enabling studies.

EA-077

STAKEHOLDER QUESTION

Which shared upstream assets, utilities, controls or suppliers could cause several buildings or services to fail together?

SENSORS AND TELEMETRY

Proposed evidence: utility, central plant, network-control, distribution and service-state telemetry from approved shared systems. Coverage: portfolio dependencies at a disclosure level appropriate to the user role. Timing: near real time for current state and quarterly topology assessment; aggregate as shared dependency groups and affected services. Quality limits: undocumented external supplies, common physical routes and correlated environmental hazards may remain hidden. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the approved portfolio dependency and utility topology, maintained by the Estates engineering, risk and Business Continuity owners. Corroboration: single-line diagrams, central plant records, utility contracts, network architecture, service maps, incident history and projects. Precedence: current approved topology and competent owner confirmation take precedence; correlation alone does not prove a common dependency. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag stale diagrams, undocumented ties, shared suppliers, temporary routes, restricted information and incomplete cross-building mappings. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: comparison and bounded diagnosis. Decision cadence: common-cause and correlated-failure screening, quarterly and after topology change. Minimum evidence: asset-to-service graph across buildings, shared nodes or routes, contingency evidence and at least one owner-validated dependency path. Method: graph cut sets, concentration metrics and incident-correlation checks with explicit non-causal caveats. Uncertainty: dependency confidence and size of potential common impact. Safe failure: return an unverified cluster and request engineering validation when the common path is inferred rather than documented.

ANSWER BOUNDARY

Do not disclose sensitive infrastructure broadly, predict simultaneous failure as certain, initiate tests or certify portfolio resilience. Accountable portfolio owners retain authority over resilience, carbon and investment pathways; scenarios are not forecasts or commitments.

WHY THIS IS USEFUL

Portfolio teams can expose concentration risk that building-by-building plans would miss.

EA-078

STAKEHOLDER QUESTION

Which adaptation actions remain useful across several plausible climate, energy-price and service-demand scenarios?

SENSORS AND TELEMETRY

Proposed evidence: temperature, rainfall or flood indicators, energy demand, asset performance and service-state data relevant to adaptation outcomes. Coverage: portfolio areas and systems within the scenario boundary. Timing: seasonal and annual aggregation with event evidence for extremes; aggregate by action and scenario. Quality limits: short local records may not represent future extremes and operational data cannot establish climate probability alone. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the approved scenario set and adaptation option definitions, maintained by the Estates Strategy, Sustainability and institutional risk owners. Corroboration: competent climate evidence, hazard assessments, asset condition, costs, dependencies, project pipeline and service requirements. Precedence: authorised scenarios and specialist evidence take precedence; analytics compares options within them. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag inconsistent horizons, double-counted benefits, uncertain costs, omitted interactions and outdated hazard assumptions. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: comparison and recommendation. Decision cadence: robust decision and regret analysis, at strategy updates and major investment gates. Minimum evidence: defined scenarios, option costs, performance criteria, dependencies and thresholds for acceptable outcome. Method: scenario matrix, dominance, worst-case regret and adaptive-pathway triggers without invented probabilities. Uncertainty: robustness across scenarios and conditions where an option fails. Safe failure: present no-regret, contingent and scenario-specific options without selecting an approval.

ANSWER BOUNDARY

Do not predict climate outcomes, approve investment, guarantee resilience or replace specialist climate, engineering and finance advice. Accountable portfolio owners retain authority over resilience, carbon and investment pathways; scenarios are not forecasts or commitments.

WHY THIS IS USEFUL

Leaders can prioritise no-regret or low-regret actions that retain value across several futures.

STAKEHOLDER QUESTION

EA-079

What phased portfolio pathways could balance carbon reduction, condition, risk, affordability and service continuity under stated assumptions?

SENSORS AND TELEMETRY

Proposed evidence: portfolio energy, condition, availability and demand indicators used to update pathway assumptions. Coverage: candidate buildings, systems and projects over the approved horizon. Timing: monthly operational refresh with annual pathway assessment; aggregate by phase, objective and dependency. Quality limits: future funding, policy, technology, service demand and delivery capacity remain uncertain. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the approved estate, carbon, condition and capital strategies, maintained by the Estates Strategy owner with Finance, Sustainability and institutional risk owners. Corroboration: project pipeline, lifecycle costs, backlog, criticality, capacity, resilience, procurement lead times and space plans. Precedence: accountable governance sets objectives, constraints and funding; analytics only explores declared pathways. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag misaligned time bases, provisional projects, optimistic delivery rates, unpriced enabling work and conflicting objective weights. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: forecast and recommendation. Decision cadence: multi-objective phased pathway analysis, annually and after major assumption change. Minimum evidence: eligible interventions, costs and ranges, dependencies, delivery capacity, carbon effects, risk or condition effects and service constraints. Method: scenario-based sequencing with Pareto frontiers, trigger points and stranded-asset checks. Uncertainty: a set of trade-off pathways and sensitivity to disputed assumptions. Safe failure: return alternative pathways and evidence gaps rather than one optimal answer when objectives or weights are not authoritative.

ANSWER BOUNDARY

Do not approve a strategy, commit budgets, certify carbon outcomes, promise delivery or create asset valuations. Accountable portfolio owners retain authority over resilience, carbon and investment pathways; scenarios are not forecasts or commitments.

WHY THIS IS USEFUL

Decision-makers can compare feasible sequences, dependencies and trade-offs before choosing a governed portfolio direction.

STAKEHOLDER QUESTION

EA-080

Which evidence gaps most limit confidence in portfolio resilience, decarbonisation and long-term asset scenarios?

SENSORS AND TELEMETRY

Proposed evidence: coverage, calibration, health, gap and mapping status for energy, condition, demand and resilience telemetry. Coverage: all datasets used in the selected portfolio scenarios, summarised without exposing restricted infrastructure. Timing: monthly data-quality refresh and at each scenario update; aggregate by evidence gap, affected decision and portfolio reach. Quality limits: instrumented assets can dominate the picture, while unknown uninstrumented systems may be more important. Evidence gate: proposed until deployment, mapping, commissioning, permission and current health are verified. Record modality/location, coverage, sampling/aggregation, units, calibration or functional check, clock alignment, expected reporting, uptime, gaps, drift and source/configuration provenance. If a decision-critical check fails, return unavailable rather than infer.

AUTHORITATIVE SUPPORTING DATA

Primary authority: the scenario evidence register and data-owner map, maintained by the Estates Strategy and Information Governance owners. Corroboration: asset records, surveys, finance, energy, risk, projects, service demand, climate evidence and permissions. Precedence: accountable data and decision owners define minimum evidence and consequence; no source is promoted because it is easy to access. Time/access/provenance: show source owner, effective or observed time, retrieval time, field-level permission and provenance. Staleness/conflicts: apply the owner's decision-specific currency threshold; flag missing owners, incompatible boundaries, stale surveys, estimated consumption, uncertain costs, hidden dependencies and conflicting horizons. Preserve unresolved conflicts for the accountable owner; never select a convenient value.

ANALYSIS REQUIRED

Operation: comparison and recommendation. Decision cadence: decision-focused evidence-gap prioritisation, at every portfolio scenario cycle. Minimum evidence: the decision or scenario output, required evidence, current source quality, uncertainty contribution, owner and feasible acquisition route. Method: uncertainty contribution and value-of-information screening with sensitivity and reach. Uncertainty: rank stability, uncertainty reduction range and unaddressable gaps. Safe failure: present an unranked set where acquisition cost, owner or decision consequence is unknown.

ANSWER BOUNDARY

Do not authorise surveys or sensing, expose restricted gaps, promise that more data will resolve strategic uncertainty or use a score to approve investment. Accountable portfolio owners retain authority over resilience, carbon and investment pathways; scenarios are not forecasts or commitments.

WHY THIS IS USEFUL

The portfolio owner can commission a proportionate evidence plan focused on gaps most likely to change a governed decision.

Using the catalogue for accountable estates decisions

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A concise path from stakeholder question to evidence-backed next action

1

Frame the decision

State the asset, space or portfolio boundary, decision date, responsible owner and permitted detail.

2

Resolve authority

Identify the current authoritative source owner and the precedence rule before comparing corroborating evidence.

3

Verify proposed evidence

Confirm deployment, mapping, commissioning, permission and current health. Unverified sensing remains proposed.

4

Check minimum evidence

Apply the record-specific minimum evidence requirement. Missing identity, time, coverage or source quality can stop the answer.

5

Run the bounded operation

Use only the named lookup, observation, calculation, estimate, forecast, comparison, diagnosis or recommendation method.

6

Expose uncertainty and conflict

Report assumptions, ranges, confidence, staleness, gaps and unresolved source conflicts in the output.

7

Apply the answer boundary

Do not convert evidence into certification, approval, authority, valuation, control action or restricted disclosure.

8

Route the next accountable action

Safe default

When evidence cannot support the decision claim, return unavailable, conflicted, stale, not authorised or verification required - with the missing evidence and accountable owner clearly stated.

Evidence, permissions and disclosure rules

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Non-negotiable controls for any implementation of the catalogue

Evidence integrity

Identity

Use stable asset, system, meter, room or project identifiers. Probable matches stay explicit.

Coverage

State what is observed and what is not. Do not generalise from instrumented assets to unobserved populations without a justified method.

Time

Carry observed/effective time, retrieval time, sampling/aggregation interval and clock quality.

Sensor quality

Record commissioning, calibration or functional checks, expected range, uptime, gaps, drift and maintenance state.

Provenance

Retain source system, field or point lineage, transformation version and configuration where relevant.

Authority and uncertainty

Precedence

Current authoritative owners and competent professionals take precedence for the decision context.

Conflicts

Preserve conflicting values and owners. Do not silently reconcile them.

Staleness

Apply owner-defined currency thresholds that are appropriate to the specific decision.

Minimum evidence

Each answer requires its stated minimum evidence. If unmet, stop safely.

Uncertainty

Show ranges, assumptions, confidence or evidence-quality qualifiers proportionate to the operation.

Permissions and boundaries

Purpose limitation

Retrieve only information needed for the authorised estates purpose and role.

Restricted assets

Do not expose sensitive plant, security, access, isolation, resilience or vulnerability detail to unauthorised users.

People and activity

Do not identify or track people, infer productivity, inspect research content, speech or network content, or infer protected characteristics.

No certification or authority

Do not approve, procure, dispose, maintain, value, certify, authorise work/access/switching, or control plant.

Historical plan

The May 2021 floor plan is historical spatial reference only; current spatial claims require a current controlled source or competent confirmation.

Implementation status: Sensors, integrations and analytics remain proposed until deployment, mapping, commissioning, permission and current health are independently verified.